

BASIC COMPUTER SKILLS 2.0 TEACHING MANUAL CLASS VII



STATE INSTITUTE OF EDUCATION

DEPARTMENT OF EDUCATION
ANDAMAN & NICOBAR ISLANDS

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MESSAGE

In today's digital age, possessing basic computer skills is no longer a luxury but a necessity. Computers have become an integral part of our personal and professional lives, opening doors to countless opportunities. From communication and research to job applications and online transactions, computer literacy empowers us to navigate the modern world with confidence.

This book on Basic Computer Skills aims to provide the students with a comprehensive understanding of essential concepts, tools, and techniques. This book will guide the students to have a step by step knowledge about the basics of Computer, even if they are a beginner. They will learn about operating systems, word processing, spreadsheets, email communication, internet browsing, and more.

By mastering these skills, they'll not only increase their efficiency and productivity but also unlock new avenues for growth and personal development. I am sure that the introduction of this manual will enable students to embrace the world of technology and empower themselves for success in the digital realm.

I believe that this Teaching Manual will serve as a pivotal moment in revolutionizing the way we educate Middle Level students about computers, and will make them stronger and more knowledgeable in the technical field.

(Pallavi Sarkar)
Secretary (Education)

FOREWORD

The Department of Education, Andaman and Nicobar Islands in collaboration with Samagra Shiksha, UT Mission Authority has been successful in providing ICT Lab facility to 129 Government schools at Middle & Secondary Level throughout the islands and 14 more schools will be added in the number taking the total to 143 schools with proper Lab facilities, Computer Systems, Internet facility and an ICT Instructor.

The students at the Middle & Secondary Level were unable to make use of these facilities to the fullest due to the non-availability of a proper curriculum and syllabus to be taught at different levels. With the objective to have a well-designed structure to introduce the Computer Aided Learning through ICT based skills, the State Institute of Education with the help of a team of Computer Teachers and subject experts have developed a Teaching Manual for the students at Middle Level (Class VI – VIII). The CBSE has also given an outline of the curriculum for IT as a Skill Subject at the Middle Level. The content of the course has also been designed as such to make students capable of independently working on a desktop and be able to develop applications to handle computations of small scale and record keeping. Students will develop the following skills: - Computational and Thinking Skills - Problem Solving Skills - Decision Making - logical and analytical skills.

This course will not only equip students with skills to gain knowledge about the basics of computer but will also instill confidence in them for developing small applications.

(Aditya Kumar Jha, DANICS)
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CHAPTER

1

FUNDAMENTALS OF COMPUTER-3

LEARNING OBJECTIVES:

At the end of this chapter, the students will be able to understand about -

- Block Diagram of Computer, Primary & Secondary Storage
- Assembling of a Computer System
- Types & Generation of Computers
- ICT & its application
- Computer Languages
- Multimedia and its use
- File types & its extensions



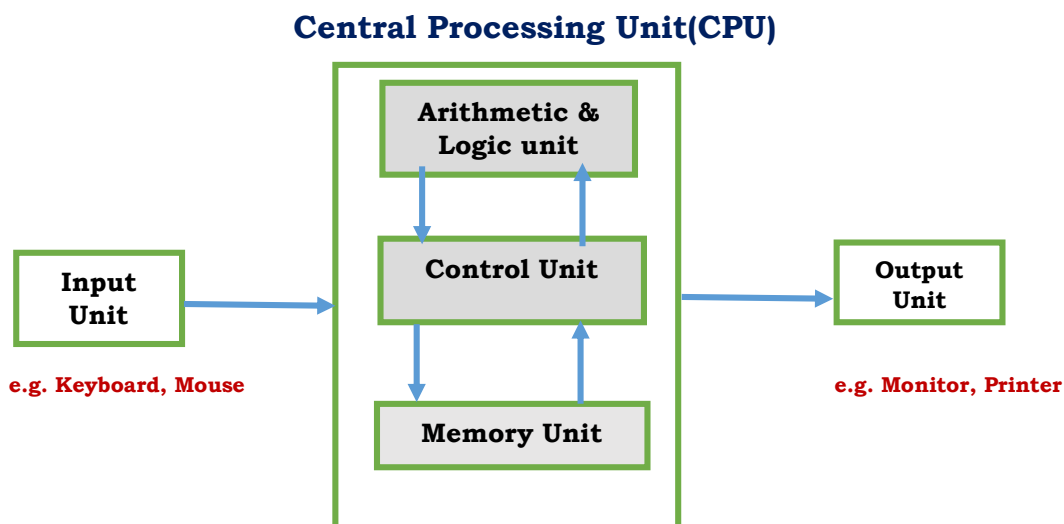
INTRODUCTION

In the previous class we have studied about the Brief History, Generation and Characteristics of Computer where the students were able to understand the basic concepts and components of Computers.

In this chapter students will be able to learn the basic structure of computer and how it works.

BLOCK DIAGRAM OF COMPUTER:

A simple computer system comprises of the basic components like Input Devices, CPU (Central Processing Unit) and Output Devices as under:





There are five basic components which include:

1. Input Unit
2. Output Unit
3. Memory Unit
4. Control Unit
5. Arithmetical and Logical Unit

➤ **Input Unit**

A computer will only respond when a command is given to the device. These commands can be given using the input unit or the input devices. For example: keyboard, mouse, joystick, scanner etc.

➤ **Output Unit**

This unit gives the result called output. It receives data from Computer System for display, physical production etc. The output can be in the form of softcopy (text, image, audio, video) or hardcopy (printed form). For example: Monitor or Printer etc.

➤ **Memory Unit**

When we enter the data into the computer using an input device, the entered information immediately gets saved in the memory unit of the Central Processing Unit (CPU). So, it consist of temporary storage area that hold the data temporarily called as Registers.

➤ **Control Unit**

This is the unit which manages the entire functioning of the computer device. It is one of the most essential components of the computer system.

The Control Unit collects the data entered using the input unit, leads it on for processing and once that is done, receives the output and presents it to the user through output unit.

➤ **Arithmetical and Logical Unit (ALU)**

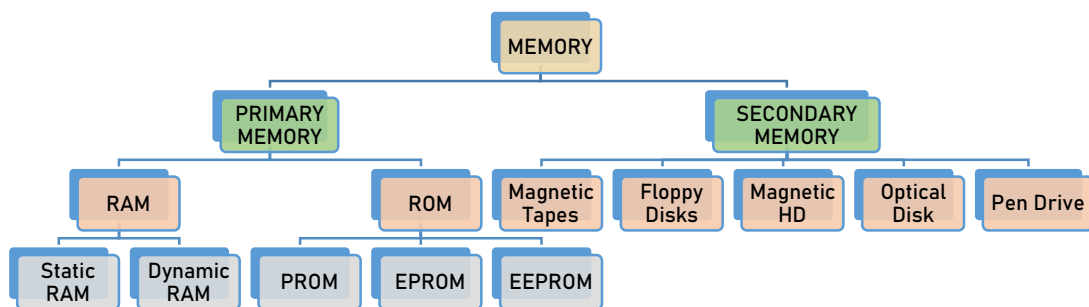
All the mathematical calculations or arithmetic operations are performed in the Arithmetic and Logical Unit of the CPU. It can also perform logical operations like comparison of data and decision-making actions.

PRIMARY & SECONDARY STORAGE (EXAMPLE AND USES):

MEMORY



Memory is a storage part in a computer system. It is used to store the data, information, and programs at the time of processing on the computer. It stores data either temporarily or permanent. The main use of memory is saving and retrieving data.



TWO TYPES OF MEMORY:

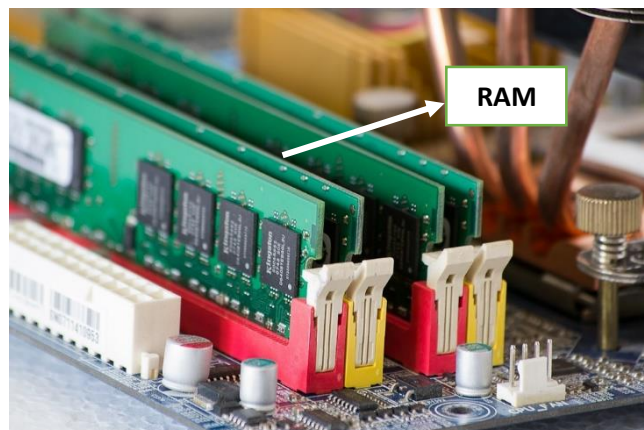


1. Primary Memory or Main Memory:

It is called the internal memory of the computer. And it is also known as main memory or Temporary memory. It holds the data and instructions that are presently working on the system or by the CPU. Primary Memory is called volatile memory, because when power is switched off it loses all data.

Primary memory is generally of two types.

- ❖ RAM
- ❖ ROM



RAM (Random Access Memory)

It stands for Random Access Memory. RAM is a read /write memory. It is referred as main memory of the computer system. It is a temporary memory. The information stored in RAM is lost whenever the power supply to the computer is switched off.



ROM (Read Only Memory)

It stands for Read Only Memory. ROM is a permanent type of memory. ROM information is not lost when power supply is switched off. The Content of ROM is inserted by the computer manufacturer and permanently stored at the time of manufacturing. ROM cannot be overwritten by the computer. It is also called **Non- Volatile Memory**.

2. Secondary Memory:

It is an external memory of the computer. It is also known as Auxiliary memory or permanent memory. It is used to store different programs and the information permanently. We call it a non-volatile memory that means the data is stored permanently even if power is switched off.

The secondary storage devices are as follows –

- Hard Disk Drive (HDD)
- Solid State Drive (SSD)
- SD Card (Secure Digital Card)
- Pen Drive
- Optical Disk (CD, DVD, Blu-Ray)
- Memory Card



ASSEMBLING OF A COMPUTER SYSTEM:

- ❖ Computer assembly is a process in which all the internal components of the computer system are fitted to make the computer functional.
- ❖ The main components involves CPU, motherboard, memory, disk drives, etc.
- ❖ There is a proper process of attachment and installation of each and every component.



- | | |
|---------|---|
| Step 1 | • First Unpack all the components (UPS, Monitor, Keyboard, Mouse, Computer Case (pre-installed with Motherboard, SMPS, CPU, CPU Fan, RAM, Network Card, HDD loaded with OS, DVD Drive) |
| Step 2 | • Place the Components on the desk or work area |
| Step 3 | • Take the UPS (Unintterrupted Power Supply) cable and plug it into main switch(wall outlet). UPS acts as a surge protector and provides temporary power if there is a power outage. |
| Step 4 | • Connect the one end of power cord cable in the computer case and the other end three pin plug in the UPS. |
| Step 5 | • Check the Keyboard for USB(rectangular) connector or a PS/2 (round) connector. If it uses a USB connector, plug it inot USB or otherwise plug it into the purple keyboard on the back of computer case. |
| Step 6 | • Check the mouse for USB or PS/2 (round) green colour port connector. Plug it into the computer case accordingly. [If the Keyboard/Mouse is wireless, a bluetooth dongle (USB connector) is needed to be attached in the computer case] |
| Step 7 | • To connect the external speakers or headphones, connect it to the green port of audion port in the front/back of the case. if it is microphone, connect it to the pink port. The blue port is the line in which MIDI devices can be used. |
| Step 8 | • Connect the Ethernet Cable (if any) in the Ethernet port and the other end in any of the network devices like MODEM, Switch, etc. |
| Step 9 | • Switch on the main plug in the wall outlet. Plug in UPS, press the power button in the computer case to start the computer system. |
| Step 10 | • Install other softwares to personalise computer. |

Note: Full Form

VGA- Visual Graphic Adapter

HDMI – High Definition Multimedia Interface

UPS – Uninterrupted Power Supply

CLASSIFICATION OF COMPUTERS

Based on Size	Based on Mechanism	Based on Purpose
• Super Computer • Mainframe • Mini Computer • Micro Computer • Desktop • Laptop • Palmtop	• Analog • Digital • Hybrid	• General Purpose • Special Purpose

1. Based on Size

Super Computer:



Supercomputers are capable to solve immense mathematical calculations, scientific problems and hence it is called number crunchers.

Mainframe Computer:

They are designed in such a way that they can support hundreds or thousands of users at the same time. Mainly used in business organisations. They are also used as servers for large data processing.



Mini Computer:

They are small in size and more compact and less expensive. They are designed for computerization of data, scientific research, industrial process, data collection and telephone switches.



Micro Computers:

The smallest among them are microcomputers. They are small in physical size most of them are desktop system.



2. Based on Mechanism(Technology)

Analog Computer:

It works on continuous data that changes in physical quantity. E.g. speedometer etc.

Digital Computer:

It works on discrete values or binary values. E.g. Digital Thermometer, Digital Watches etc.

Hybrid Computer:

It is a combination of analog and digital computer

3. Based on Purpose

General Purpose Computer:

These are designed and developed to perform day-to-day operations. E.g. Personal Computer, Tablet etc.

Special Purpose Computer:

These are designed to perform specific task. E.g. ATM, Scientific Research Computer, Rocket and Space station Computer etc.

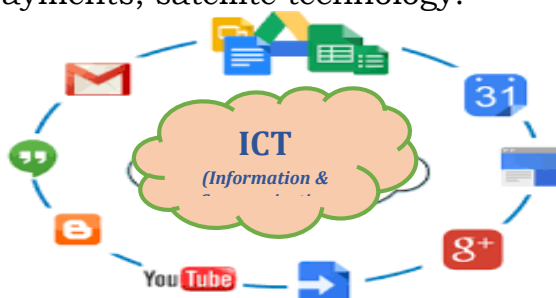
ICT (Information and Communication Technology)

- ❖ ICT stands for Information and Communication Technology. It includes a diverse set of technological tools and resources used to communicate, create, disseminate, store, and manage information. ICT includes computers, the Internet, broadcasting technologies (radio and television) and telephony.
- ❖ People use technology every day at home and at workplace. ICT tools enable anytime, anywhere access to information and resources. The various ICT tools that are used include computers we use at work, smartphones through which we talk, Internet for getting information, communicating with others through e-mail and social networks, watching movies and TV shows, etc.



Application of ICT

- ❖ ICT is used in most of the fields such as E-Commerce, E-governance, Banking, Agriculture, Education, Medicine, Defense, Transport, etc. making daily life much easier.
- ❖ Examples of ICT include computers, mobile phones, the internet, robotics, artificial intelligence, digital television, video conferencing, online banking and payments, satellite technology.



GENERATION OF COMPUTERS:

Computer Generation	Features	Image
First Generation	➤ They used Vacuum Tubes and Magnetic Drums for the Memory. ➤ They occupied a lot of Space. ➤ They were Expensive to Operate. ➤ They used lots of Electricity and generated lots of Heat. ➤ Example: ENIAC, EDVAC, EDSAC [ENIAC: Electronic Numerical Integrator and Computer]	
Second Generation	➤ They used Transistors and Magnetic Core Memory. ➤ They were Smaller and Faster compared to First-Generation Computers. ➤ Examples: IBM 1401, IBM 700, UNIVAC III.	
Third Generation	➤ They used Integrated Circuits (ICs). ➤ They were faster and efficient than second – generation Computers. ➤ They used Operating System, allowing many different applications to run at one time. Example: IBM 360, ICL 1900.	
Fourth Generation	➤ They used VLSIC (Very Large Scale Integrated Circuit) and ULSIC (Ultra Large Scale Integrated Circuit). ➤ They are Smaller, Energy Efficient and more Reliable than the earlier Generation of Computers ➤ Examples: Desktop.	
Fifth Generation	➤ These Computers will be based on Artificial Intelligence. ➤ Intelligence is the Branch of Computer Science that deals with making the Computers think and take decisions like Human beings.	



DID YOU KNOW??

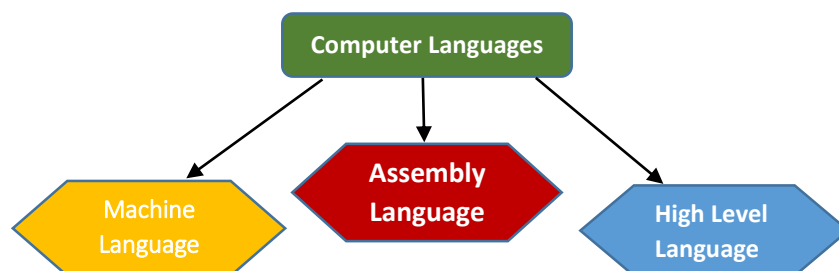
Google receives more than 99,000 searches every second



Computer Languages:

A Language is a Medium of Communication through which the user and the Computer communicates with each other.

- A computer language can be classified into:



1. **Machine Language:** The languages that are written by using machine codes i.e. binary digit 0's and 1's are known as machine language or low level.

Example of machine language for the text "Hello World": -

01001000 01100101 01101100 01101100 01101111 00100000
01010111 01101111 01110010 01101100 01100100

2. **Assembly Language:** An assembly language statement is a line of text that translates into a single machine instruction.
3. **High Level Language:** The language that are written by using normal English instruction is known as high level language.

Eg:- Java, PHP, C, C++ etc.



Multimedia and its use:



Multimedia means that computer information can be represented through audio, video, and animation in addition to traditional media (i.e., text, graphics drawings, images).



Multimedia is used in many fields. Some of the important application areas of multimedia are given below.

1. Multimedia is used in the field of education.
2. Multimedia is used to represent the model of human body in medical science.
3. Multimedia is used in the field of entertainment.
4. It is used to inform customers at railway stations, banks etc.
5. It is used for face to face communication with people living far away.

FILE TYPES & ITS EXTENSIONS:

File is collection of meaningful information, pictures and objects. There are various types of file, such as text file, picture file and sound file. For example, notepad file will be collection of text, an image file, such as -gif, Jpeg file is collection of picture. Shortcut for text file is .txt.

Types of Files

-  JPEG
-  PNG
-  GIF
-  PDF
-  SVG
-  MP4



JPEG (Joint Photographic Experts Group)

- JPEG is the most common format used by digital cameras and images living on the World Wide Web.

PNG (Portable Network Graphics)

- PNG is a high-quality file format used for images.

GIF (Graphics Interchange Format)

- The GIF file is a form of bitmap images, meaning the graphic is composed of many tiny parts called pixels, just like the JPEG and PNG file formats.

PDF (Portable Document Format)

- This file format is used for online documents and printing purposes.

SVG (Scalable Vector Graphics)

- SVG is a format commonly used to save and display vectors.

MP4 (Moving Picture Experts Group)

- MP4 is a multimedia container format, which is just a fancy term for most of the videos we consume on the Internet.

ASSESSMENT

Fill in the blanks:

- The Unit where all the processing is done is called as
- Keyboard, mouse, joystick, scanner are the examples of
- is a storage part of a computer system.

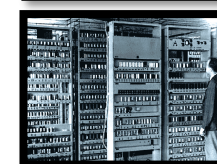
4. Auxiliary memory is also known as.....
5. is collection of meaningful information, pictures and objects.

Expand the Following

1. RAM	2. ROM	3. ICT	4. JPEG
5. GIF	6. PDF	7. PROM	8. EPROM
9. EEPROM	10. PNG		

Match the following:

1. First Generation Computer
2. Second Generation Computer
3. Third Generation Computer
4. Fourth Generation Computer
5. Fifth Generation Computer



CHAPTER

2

WORD PROCESSING (MS-WORD)

LEARNING OBJECTIVES:

At the end of this chapter, the students will be able to -

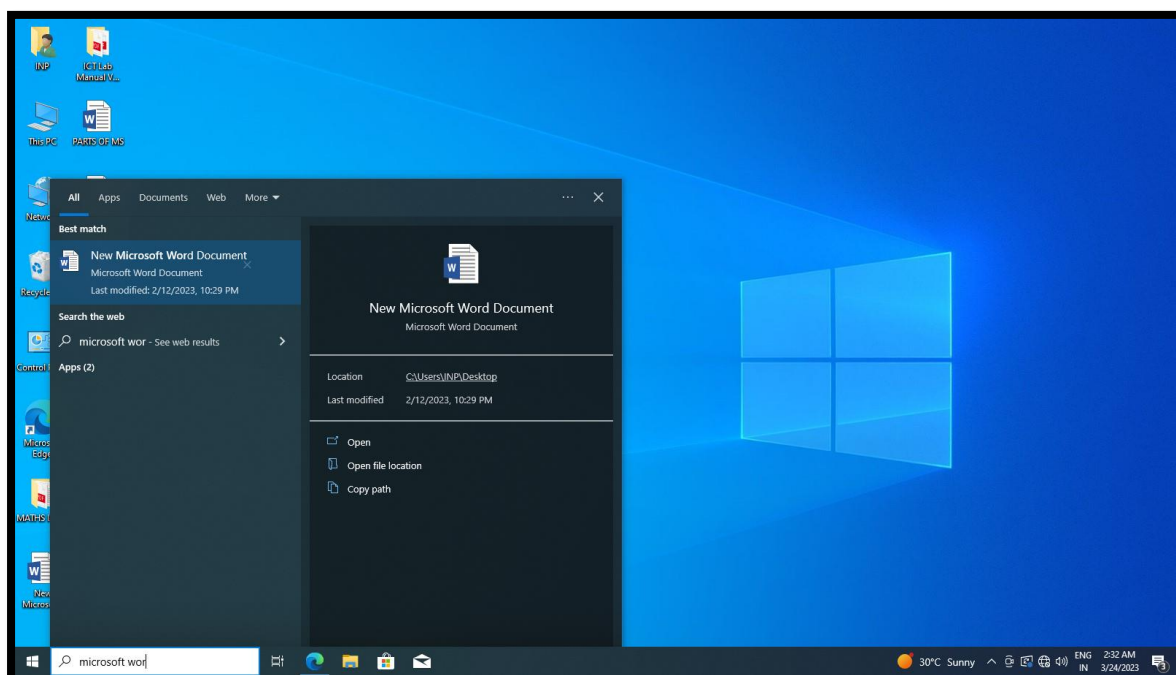
- ❖ Apply various formatting options
- ❖ Use bullets and numbering
- ❖ Insert pictures, shapes & symbols
- ❖ Apply header and footer in a document
- ❖ Create and manipulate a table
- ❖ Check spelling and grammar
- ❖ Apply basic formatting on text
- ❖ Use MS-Word shortcuts



REVIEW OF WORD PROCESSING

Activity 1 : Load MS Word window from Windows

Start → All Programs → Microsoft Office → Microsoft Word

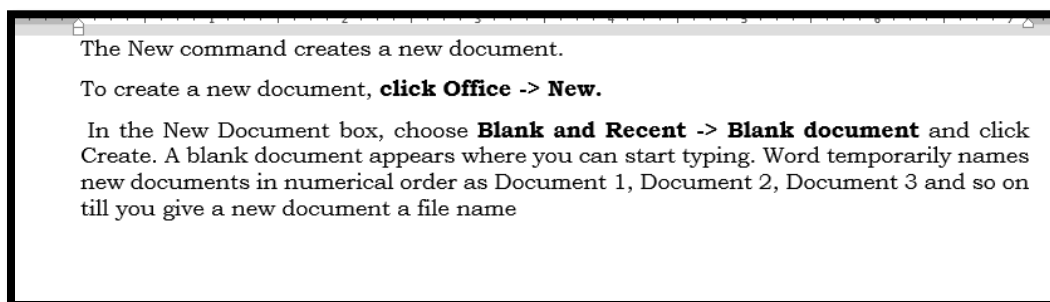


Activity 2:

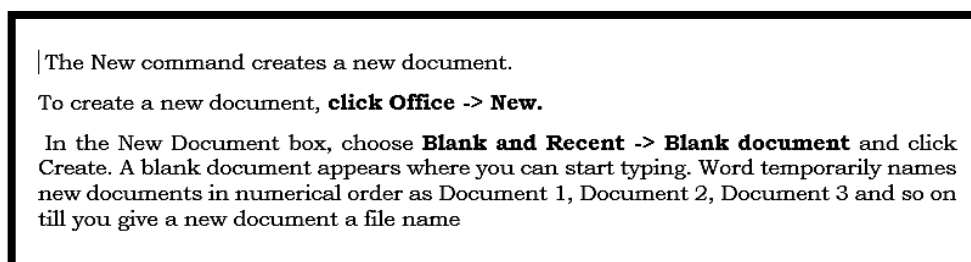
Recall the following Navigation keys.

- Down/Up, Right/Left arrow keys, End/Home keys.
- Ctrl + Home/End

Open the document called KiteWithoutThread.docx

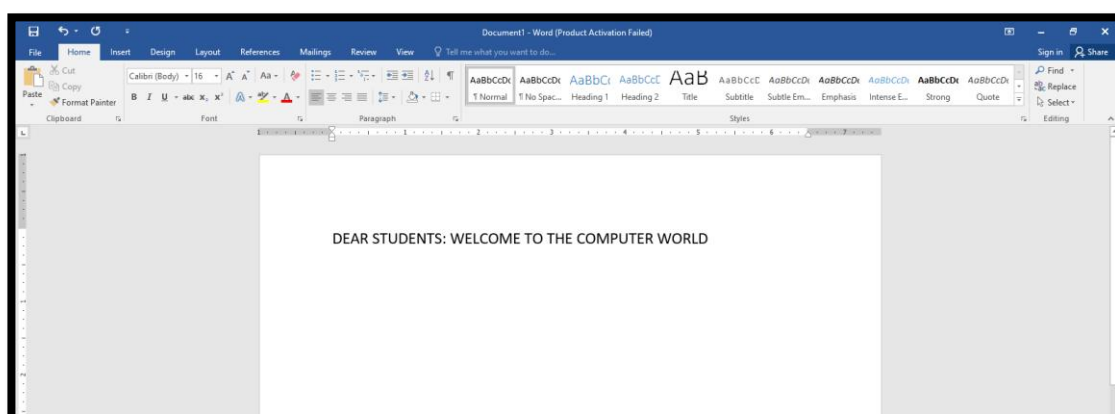


- Press Ctrl + End keys together to go to the end of the document.
- Press Ctrl + Home keys together to go to the beginning of the document.

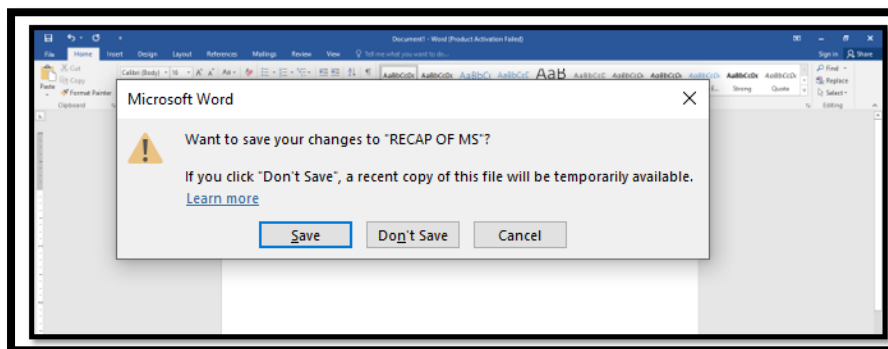


- Close the Word window by quitting Word. (File → Exit)

Activity 3a: Create a new blank document and type the following in it.



Activity 3b: Close the Word window and click Don't Save when Word shows the following



Activity 4: Type the following in a blank document.

- (a) In the beginning of the poem, write the heading **“WAITING”**. (Use the arrow key to go to the beginning of the first line, type the title and press the Enter key twice.)

Waiting

The song I came to sing
remains unsung to this day.
I have spent my days in stringing
and in unstringing my instrument

- (b) After the last line, type the name of the poet “- Rabindranath Tagore”. (Go to the end of the last line, press Enter key twice and type the poet’s name

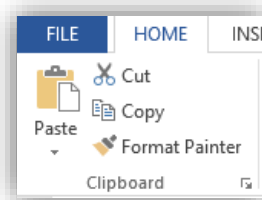
The time has not come true,
the words have not been rightly set;
only there is the agony
of wishing in my heart...

-Rabindranath Tagore

- (c) Select the entire text and press the space bar key. What happens? (Use Ctrl + A)
- (d) Select the first line “The song”

The song I came to sing
remains unsung to this day.
I have spent my days in stringing
and in unstringing my instrument

- (e) Click the **Copy(Ctrl+C)** or **Cut(Ctrl +X)** button on the **Home** tab to perform cut/copy operation.



- (f) **Paste(Ctrl+V)** it at the end of the document, below the poet's name. (Use the arrow keys to move to the last line, press **Enter** key if necessary and click the **Paste** button on the **Home** tab)

FORMATTING

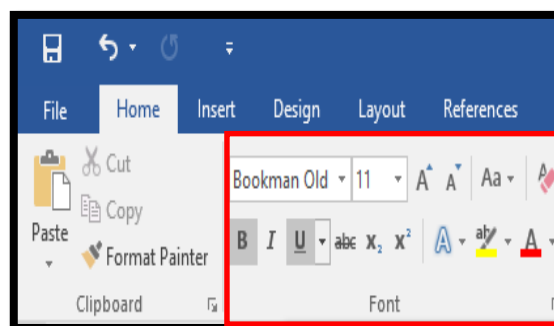
Formatting means changing the appearance of the document to make it look attractive. This can be done by changing the font type, font size, font color, text alignment, adjusting margins and bold facing, italicizing, underlining the text, etc.

There are two types of formatting in MS-Word.

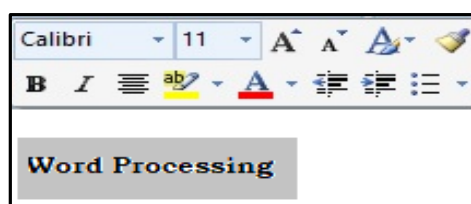
- **Character Formatting** : It is applicable to the selected text.
- **Paragraph Formatting** : It is applicable to the entire paragraph.

Character Formatting:

It is applicable to the entire paragraph. There are several ways in which we can do Character formatting. To use any of the Character formatting options available, the text has to be selected first and then only any change can be

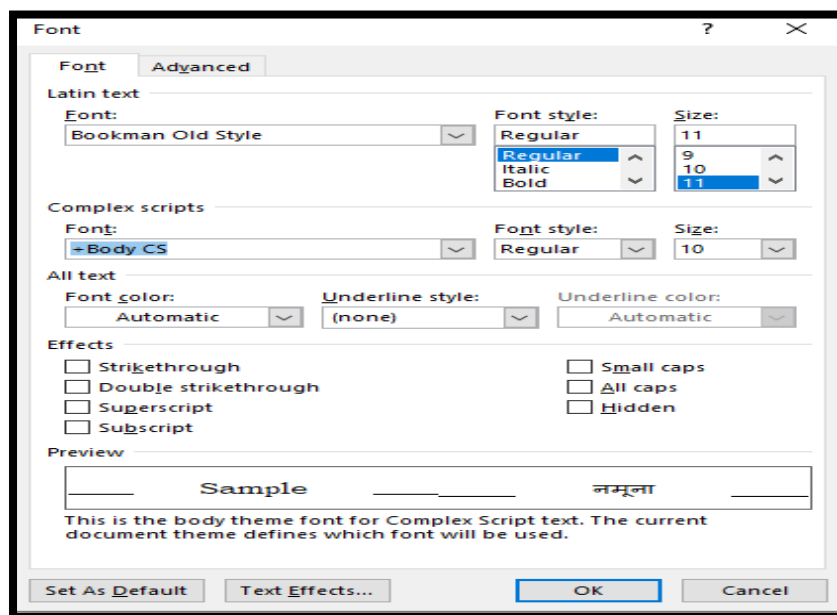


made to it. The Font group of the Home tab has many options to format the text in a document. When we move the mouse pointer over the selected text or click on it, a mini Toolbar appears with some options to format the text as shown:



Now, click on the Font dialog box launcher in the Font group. A Font Dialog box appears as shown below. The box has many options to format the text of a document.

Change the font, font style, size, colour and click on OK button to apply the effects on the selected text.



ALIGNMENT OF TEXT

The alignment of text means the manner in which the text is placed between the left and the right margin of a page. There are many ways to set the alignment as shown in the figure.

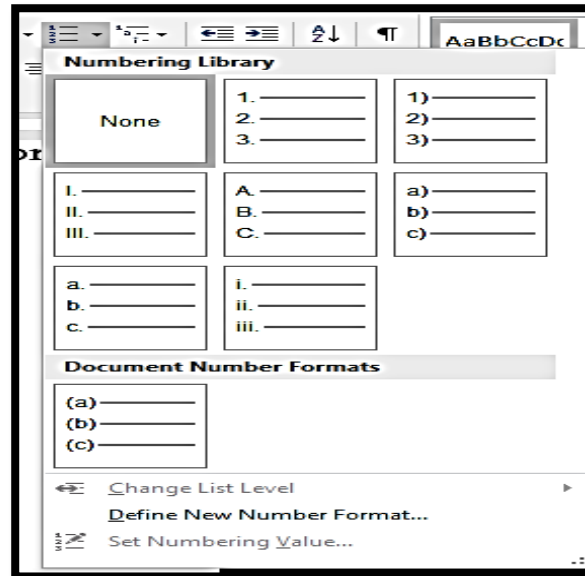
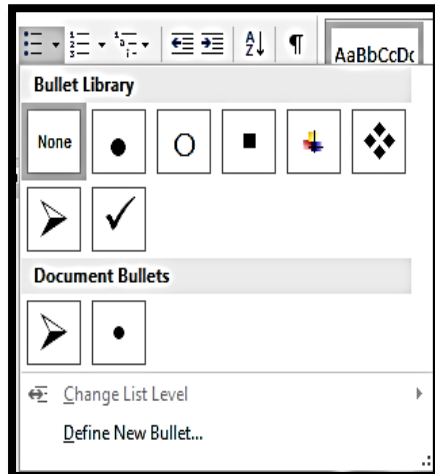
ALIGNMENT	DESCRIPTION
 Align Left	The text is aligned along the left margin
 Align Right	The text is aligned along the right margin
 Center	The text is placed between the left and right margin
 Justify	The text is evenly aligned from both left and right margin

BULLETS AND NUMBERING

You can add bullets or numbering to your text to set it in a point-wise manner. When you turn the bullets and numbering feature on, every new line begins with a bullet or a number each time you press Enter. This feature is very useful when you are making a list or a point-wise report.

To add bullets or numbering, click on  to add bullets, or click on  to add numbering to your list. To choose the style of bullets or numbering, just click on the arrow next to the Bullets or Numbering icon from the Paragraph group in the Home Tab.

To remove the bullets, select the text from which you want to remove the bullets or Numbering, and click on bullets or numbering.

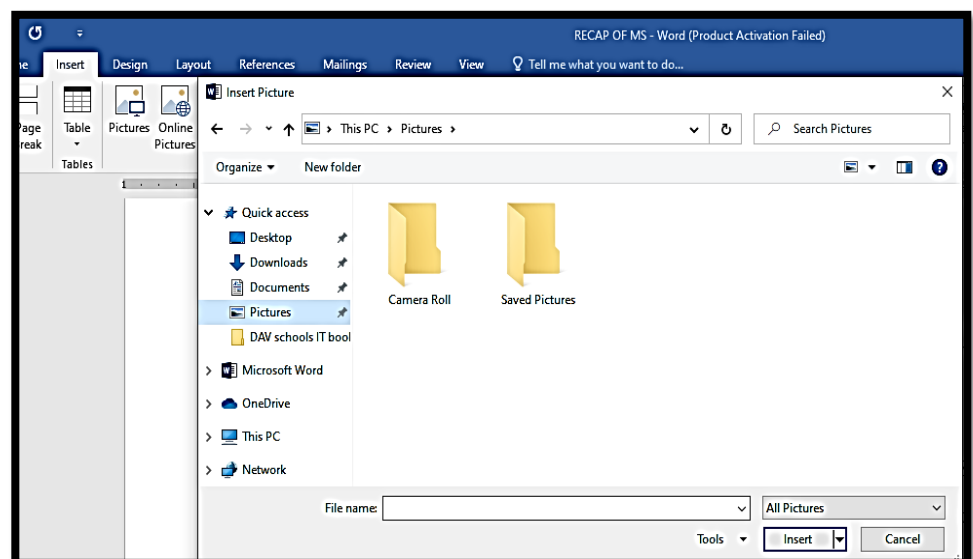


INSERTING A PICTURE

Inserting a picture in a Word document makes it more interesting and highlights points raised in the text. There are built-in clip art images for your document. If you have a more specific image in mind, you can insert a picture from a file.

Insert a picture from a File

1. Place your insertion point where you want the image to appear. Select the Insert tab.
2. Click the Picture command in the Illustrations group. The Insert Picture dialog box appears.



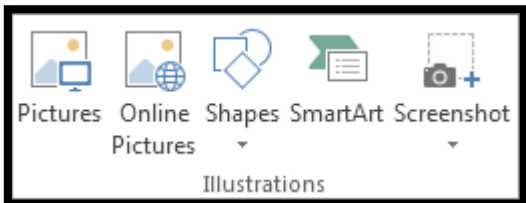
NOTE:  To resize an image, click and drag one of the corner sizing handles. The image will change size while keeping the same proportions. If you want to stretch it horizontally or vertically, you can use the side sizing handles.

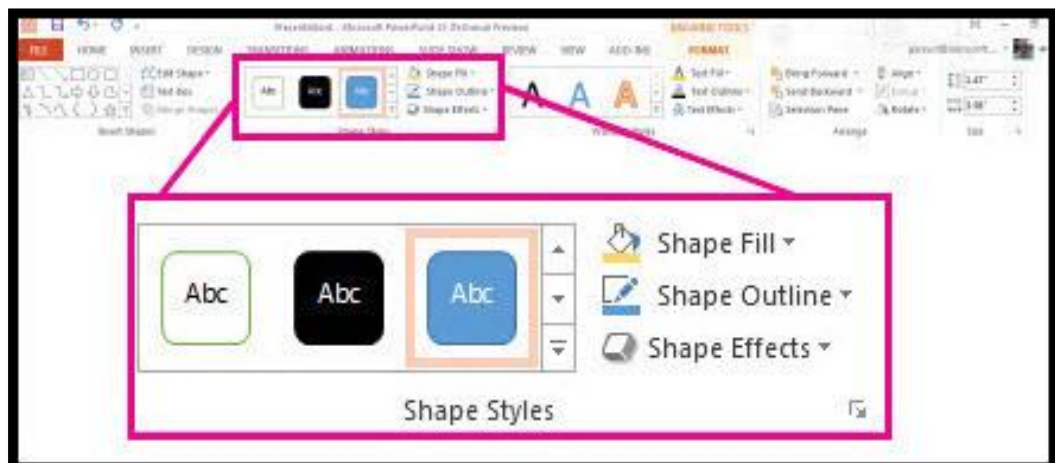
INSERT SHAPES

You can add shapes, such as boxes, circles, and arrows, to your documents, email messages, slide shows, and spreadsheets. To add a shape, click Insert, click Shapes, select a shape, and then click and drag to draw the shape.

After you add one or more shapes, you can add text, bullets, and numbering to them, and you can change their fill, outline, and other effects on the Format tab.

Steps to insert shapes

1. Click in your document where you want to create the drawing.
 2. On the Insert tab, in the Illustrations group, click Shapes.
- 
3. You can do any of the following on the Format tab, which appears after you insert a drawing shape:
 - **Insert a shape:** On the Format tab, in the Insert Shapes group, click a shape, and then click somewhere in the document.
 - **Change a shape:** Click the shape you want to change. On the Format tab, in the Insert Shapes group, click Edit Shape, point to Change Shape, and then choose a different shape.
 - **Add text to a shape:** Click the shape you where you want text, and then type.
 - **Adjust the size of the shapes.** Select the shape or shapes you want to resize. On the **Format** tab, in the **Size** group, click the arrows or type new dimensions in the **Height** and **Width** boxes.
 - **Apply a style to a shape.** In the Shape Styles group, rest your pointer over a style to see what your shape will look like when you apply that style. Click the style to apply it. Or, click **Shape Fill** or **Shape Outline** and select the options that you want.



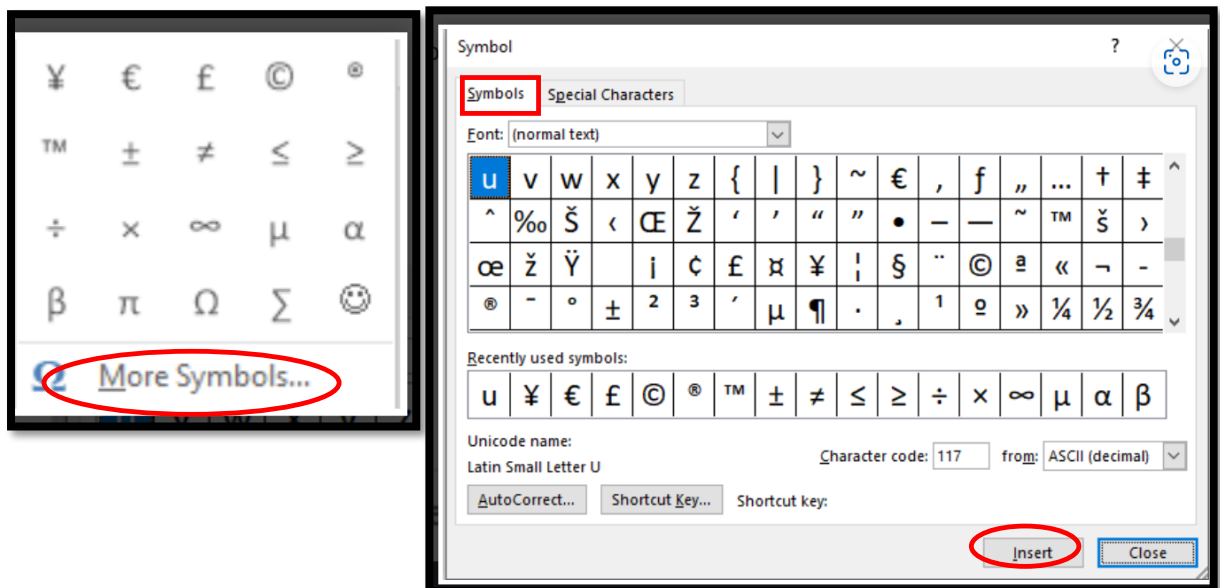
INSERT SYMBOL

Symbols in computer word processing denote data types or characters that are unique and in a human-readable form.

Microsoft Word has a special features called symbols in which user can insert many different types of symbols in the document.

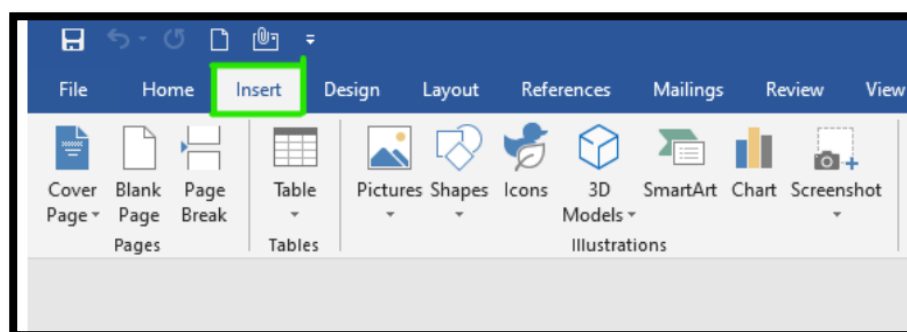
MS Word has a variety of symbols, special characters, fractions, languages, etc. We can insert the symbols and special characters with the help of the Symbol dialog box present in MS Word.

Some pictures of symbols in MS Word are:

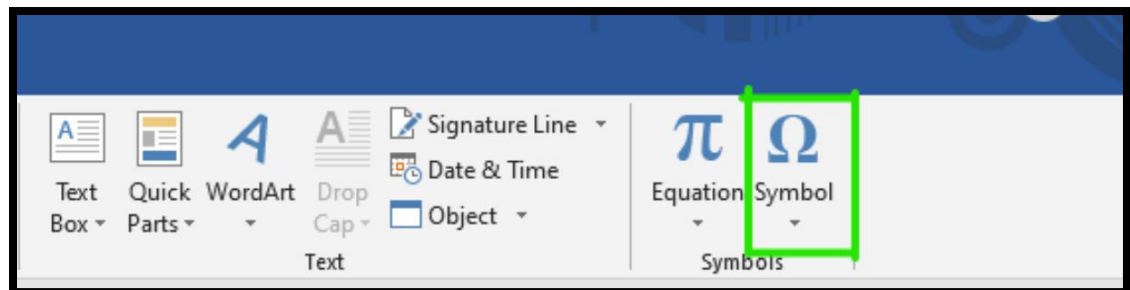


Steps to insert Symbols:

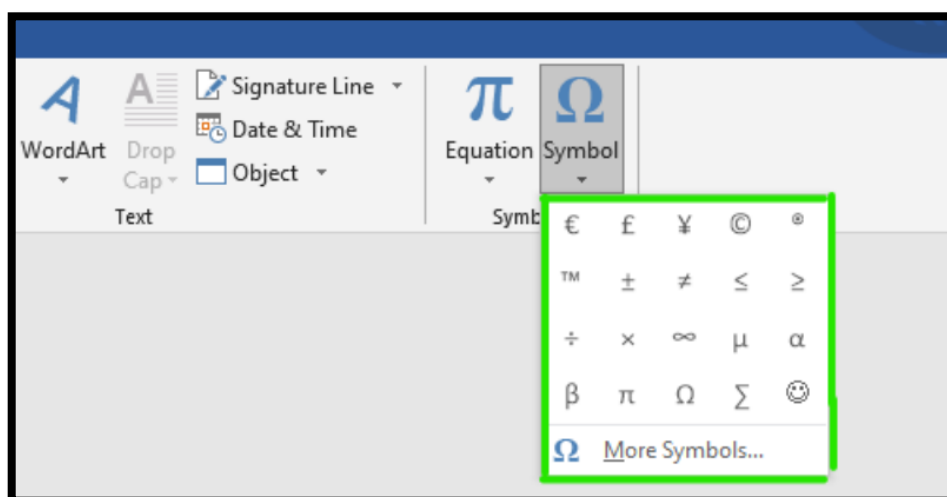
Step 1: On the navigation menu click on the **insert** option.



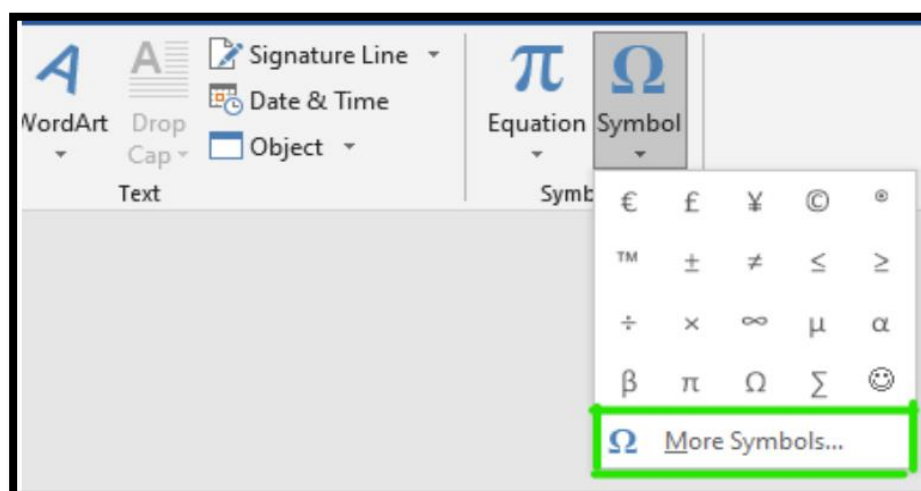
Step 2: On the right-hand side of the navigation menu select the symbol option as shown in figure:



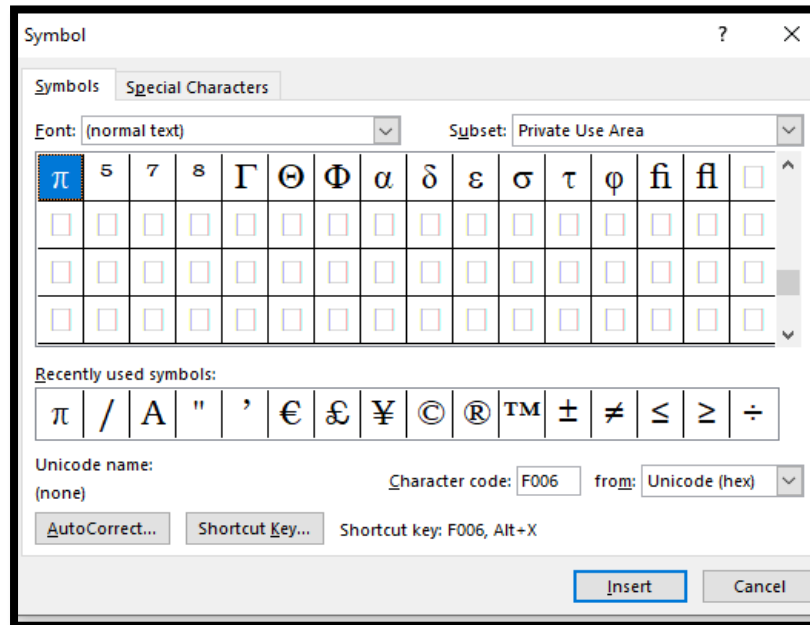
Step 3: Next select the **symbol** from the menu as per your need



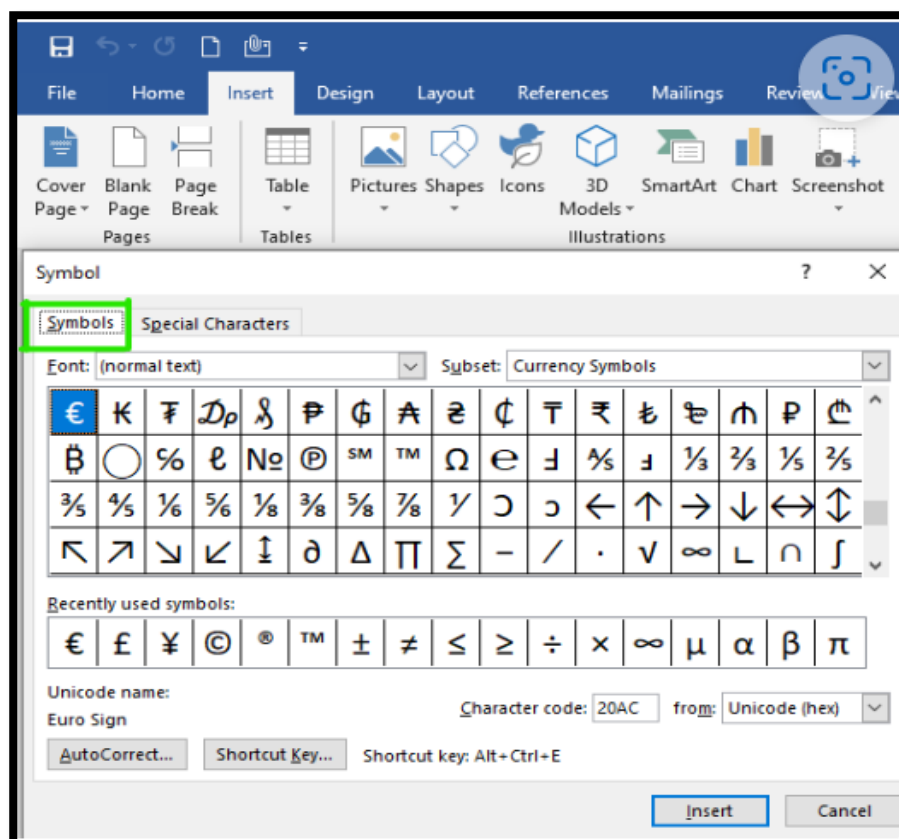
Step 4: You can also select more different types of symbols by just clicking on the **More Symbols** option as shown in the figure:



Step 5: A **symbol** dialog box will be pop up on the screen.



Step 6: Now click on the **symbols** option as shown in the figure:

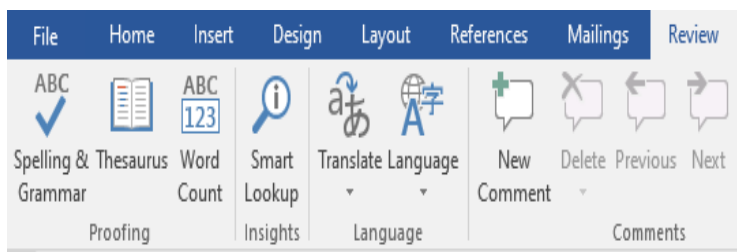


Step 7: Click on the symbol to select it.

Step 8: Click on Insert button at the bottom of dialog box.

CHECKING SPELLING AND GRAMMAR

In MS Word, there is a built-in dictionary and a set of grammatical rules to check a document. An incorrect spelling of a word in a word document can be seen with a single red wavy line beneath it. Similarly, a line with grammatical mistakes are seen with a green wavy line.



The various steps to start Spelling and Grammar checking are given below.

1. Place the cursor on the word with the spelling mistake. Click on the 'Review' tab to select 'Spelling and Grammar' button.
2. 'Spelling and Grammar' dialog box appears. The text which is misspelt is highlighted and a list of words with the correct spelling is displayed in the 'Suggestions Panel' at the bottom of the dialog box. Choose the options from the table to move further.

Shortcut : F7

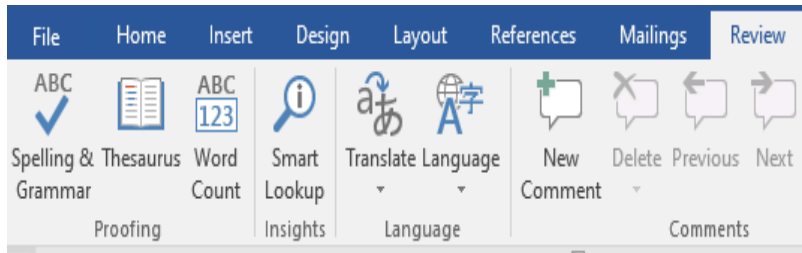
OPTION	DESCRIPTION
Ignore once	Click on this button to ignore the error shown.
Ignore All	Click on this button to ignore a particular error in the whole document.
Add to Dictionary	Click on this button to add the highlighted word to MS Word dictionary
Change	First choose a word from the Suggestions panel, then click on this button to insert the corrected word.
Change All	Click on this button to change the word wherever it appears in the document.
Auto Correct	Click on a suggestion to accept it and then click on the Change button. This will change the word on its own while we are typing.

THESAURUS

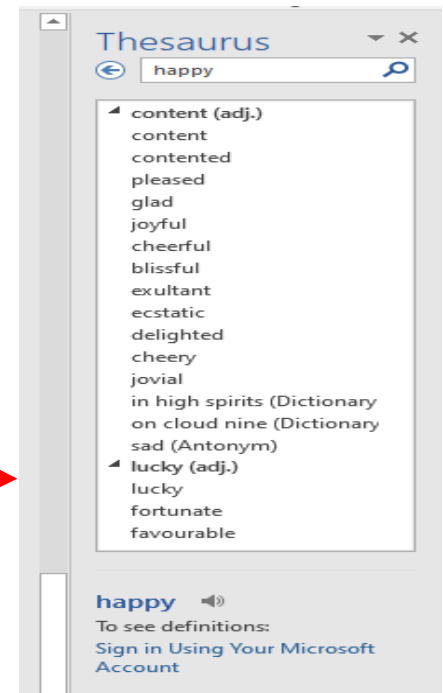
Thesaurus is a tool, which is used in MS Word to search for synonyms (different words with the same meaning). It helps writers to find exact and suitable words to express their idea.

Try → Shift+F7

Click : Review tab → Thesaurus

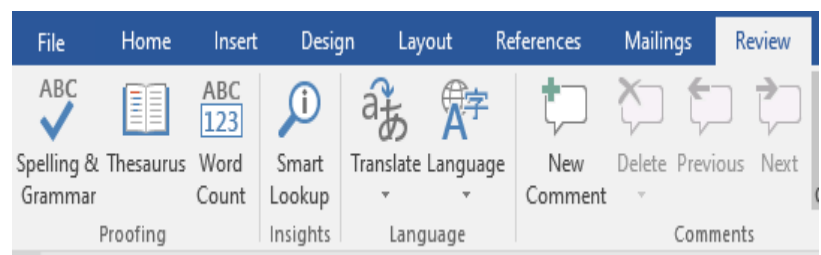


You will get screen like this :



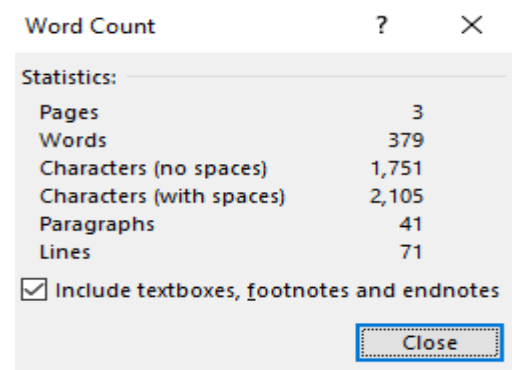
WORD COUNT

Word counts the number of words, pages, paragraphs, lines, and characters in a document while you type.



Click : Review tab → Word Count

You will get screen like this :

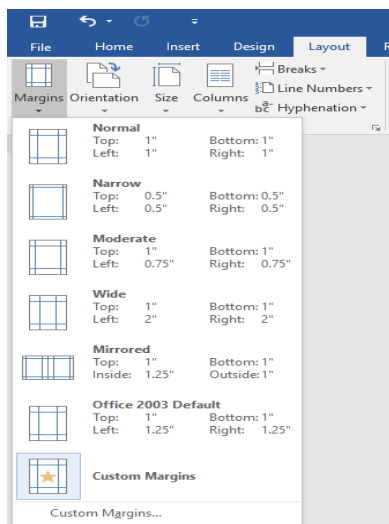
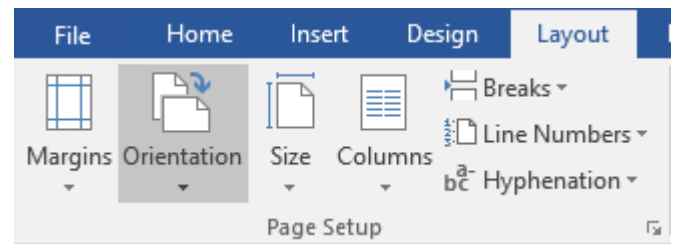


PAGE SETUP

The page setup are the parameters defined by the user that help determine how a printed page appear. Those parameters can include everything from the size, margins, page orientation, to quality of print.

MARGINS SETTING

Page margins are the blank space around the edges of the page (Top, Bottom, Left, and Right).

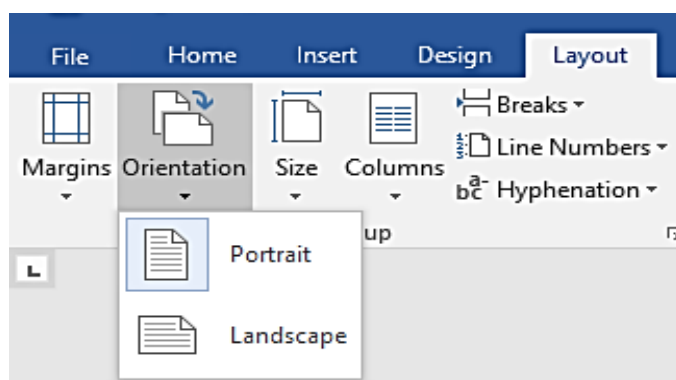


Click : Layout tab → Margins → Select the margin type that you want or specify your own margin settings by clicking on Custom Margins, and then in the Top, Bottom, Left, and Right boxes, enter new values for the margins

ORIENTATION

You can choose either portrait (vertical) or landscape (horizontal) orientation for all or part of your document.

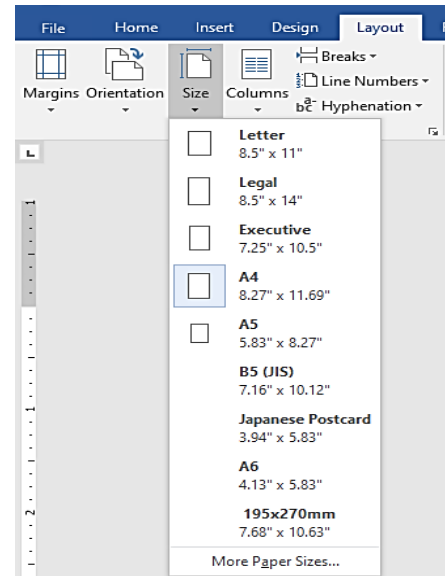
Click : Layout tab → Orientation → Select the orientation



PAPER SIZE

Page size refers to the area of your Paper or sheet, size is the size of the paper used for printing.

Click : Layout tab → Size → Select the size like Legal/A4/Letter,

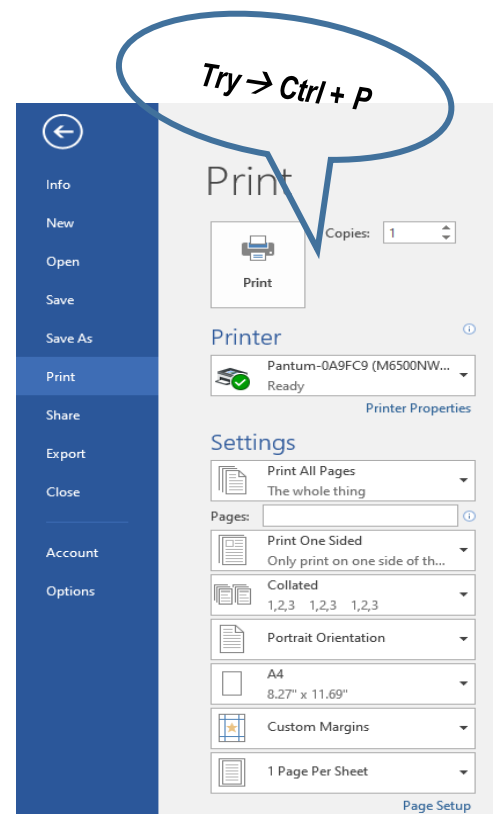


PRINTING A DOCUMENT

In MS-Word the printing commands allow you to control how the document will be printed, where it will be printed and what will be printed.

In MS-Word 2016, to print a document follow these steps :

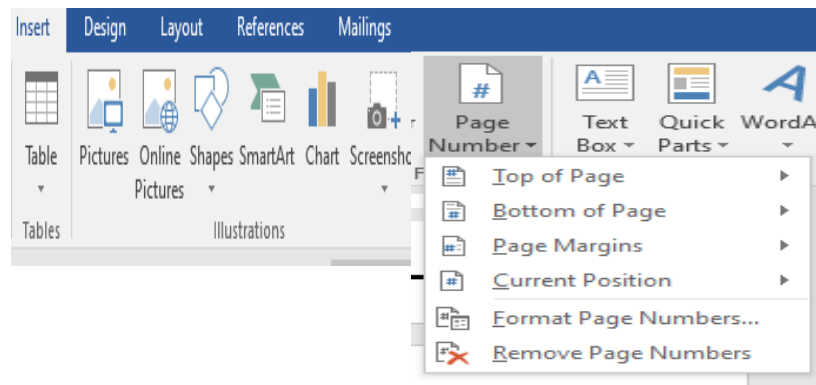
Click : File → Print → Select the printer and apply necessary page settings, if required.



INSERTING A PAGE NUMBER

Page numbers can be used to automatically number each page in your document. They come in a wide range of number formats and can be customized to suit your needs. Page numbers are usually placed in the header, footer, or side margin.

Click : Insert → Page Number → Select the place where you want to give the page number



HEADER OR FOOTER

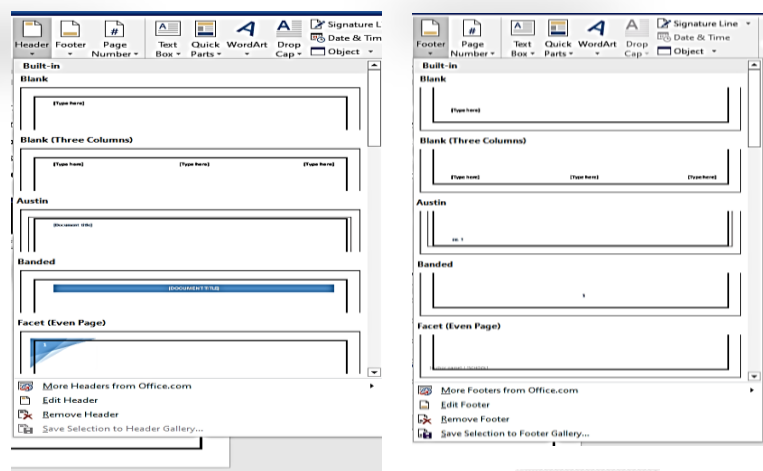
Headers are portions of a document that always appear at the top of a page; footers appear at the bottom of a page. Headers and footers generally contain additional information such as **page numbers, dates, an author's name,** and **footnotes**, which can help keep longer documents organized and make them easier to read. Text entered in the header or footer will appear on **each page** of the document.

In MS-Word 2016 -

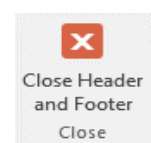
Click : Insert



A drop-down submenu opens up with a list of headers/footers. You can scroll down to get many options.



When you have finished editing, click Close Header and Footer

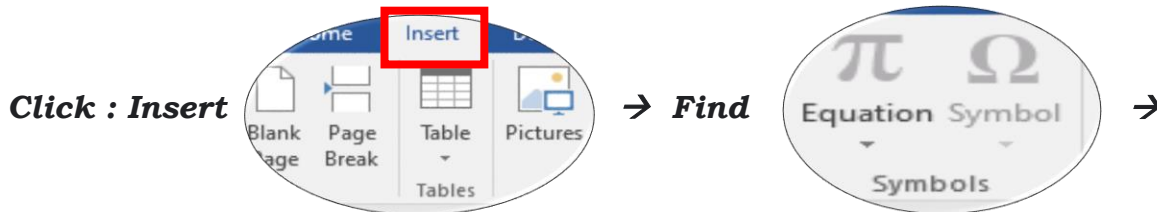


Note : After you close the header or footer, it will still be visible, but it will be locked. To edit it again, just double-click anywhere on the header or footer margin and it will become unlocked for editing.

INSERTING MATHEMATICAL EQUATIONS AND SYMBOLS

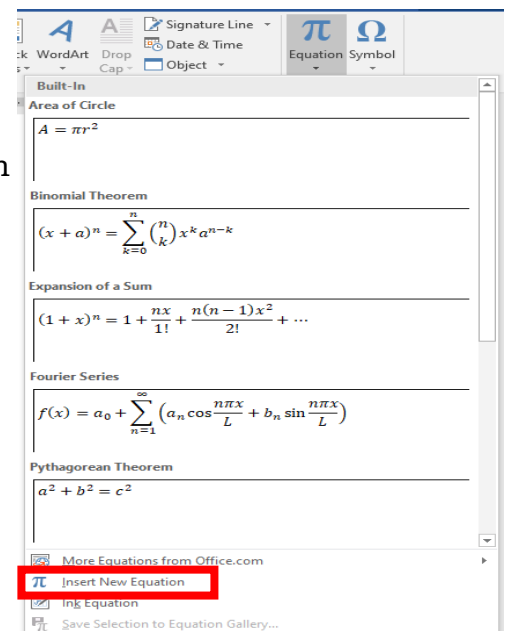
In Word, you can insert mathematical symbols into equations or text by using the equation tools. You can add symbols that are not on your keyboard. Microsoft Word includes an Equation Editor which can be used to insert common mathematical expressions in the document.

In MS-Word 2016 –



Select the equation from Built-in OR

select “Insert New Equation” to type your own equation



TABLES

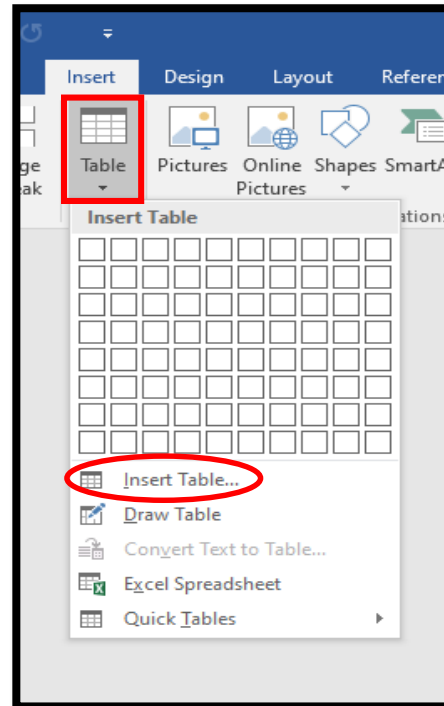
A table is a grid of rows and columns that you can fill with text and graphics.

A row is a horizontal series of cells whereas a column is a vertical series of cells. A cell is a rectangular box which is formed by the intersection of rows and columns.

Tables are used to organize and present information in a systematic manner. You can also use tables to input numbers and then perform calculations on them.

Creating a Table

- To create a table in MS Word we first open the Word document, and place the cursor where we want to insert the table.
- Click on the Insert tab.
- Click on the drop-down arrow in the Tables group.
- A drop-down menu opens up.
- Click on the grid and drag the cursor to select the number of columns and rows you want.
- The table gets inserted.

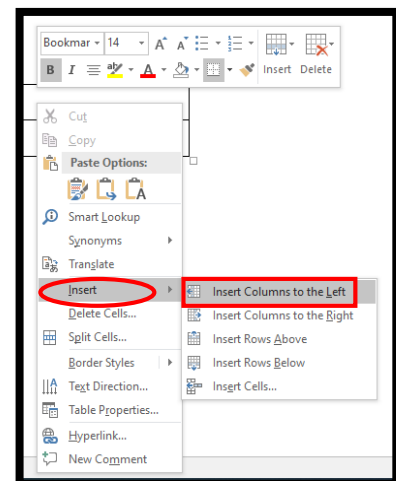


Inserting rows and columns in a Table:

Insert option in the Rows and Columns group.

A submenu opens up.

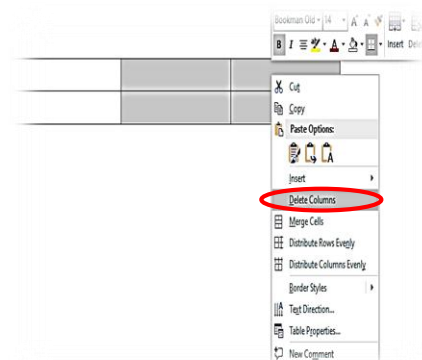
- Click on Insert Cells to the selected cell.
- Click on Insert Columns to right or left to the Selected column.
- Click on Insert Row to above or below to the selected column.



Deleting rows and columns in a Table:

To delete a table, click on the table inserted. Click on the Layout tab from the ribbon and click on Delete option in the Rows and Columns group. A submenu opens up.

- Click on Delete Cells to delete the selected cell.
- Click on Delete Columns to delete the selected column.
- Click on Delete Rows to delete the selected row.
- Click on Delete Table to delete the selected table.

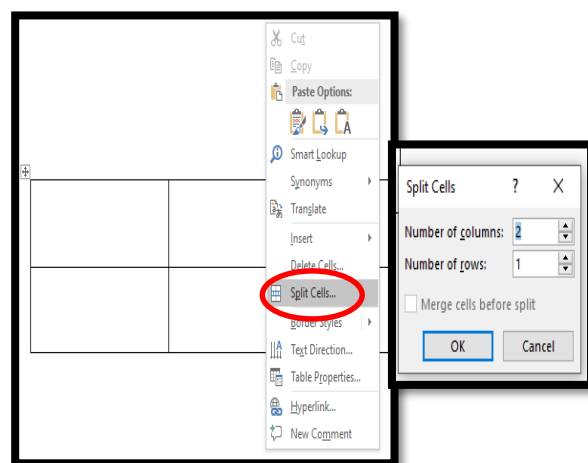


SPLITTING AND MERGING CELLS:

Split Cells:

Splitting cells is similar to adding a row or column, but it all takes place in one cell instead of a group of cells.

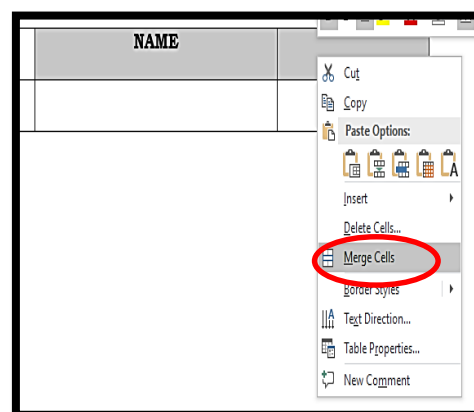
1. Select the cell that you want to split.
2. Select Layout > Split Cells.
3. Enter the number of columns or Rows that you want to split the selected cell into, and then select OK.



Merge Cells:

Merge cell is a facility that allows Multiple adjacent cells to be combined into a single larger cell. This is done by selecting all cells to be merged and choosing the **“Merge Cells” command**.

1. In the table, drag the pointer across the cells that you want to merge.
2. On the ribbon, select the Layout tab.
3. Select Merge Cells.



Copying and Moving a Table:

When you paste a table in a new location, you can copy the table or cut it. When you copy a table, the original table remains in place. When you cut a table, the original table is deleted.

1. Click the table move handle to select the table.
2. Do one of the following:
 - To copy the table, press CTRL+C.
 - To cut the table, press CTRL+X.
3. Place the cursor where you want the new table.
4. Press CTRL+V to paste the table in the new location.

MS-WORD SHORTCUT KEYS	
PARTICULARS	KEYS
Cut the selected items	Ctrl + X
Copy the selected items	Ctrl + C
Paste the selected items	Ctrl + V
Save the selected items	Ctrl + S
Undo the Task	Ctrl + Z
Redo the Task	Ctrl + Y
Go to the beginning of the document	Ctrl + Home
Go to the end of the document	Ctrl + End
Select All Text	Ctrl + A



Test your knowledge

I) Fill in the blanks:

- Ready-made gallery of pictures in MS Word is called
- is a text-styling tool of MS Word.
- To remove bullets, click button.
- Bullets and Numbering are found in the tab.
- Grammatical errors are indicated withwavy lines.
- MS Word has a built-in to check the spelling of a word.

II) Match the columns

Column A

- Delete
- Table
- Rows
- Columns

Column B

- Horizontal series of cells
- Layout tab
- Vertical series of cells
- Insert tab

III) USAGE OF SHORT CUT KEYS

- Cut the text _____
- Paste the text _____
- Select the entire document _____
- Copy the text _____
- Save the document _____

CHAPTER 3

TUX PAINT



Learning Objectives:

At the end of this chapter, the students will be able to understand about -

- Introduction of Tux Paint.
- Getting familiar with Drawing Tools.
- Learn to draw picture using Paint Tool.
- Drawing picture using Line and Eraser Tool.
- Understand use of Shape Tool.
- Getting familiar with Text and Rubber Stamp Tool, Magic Tool

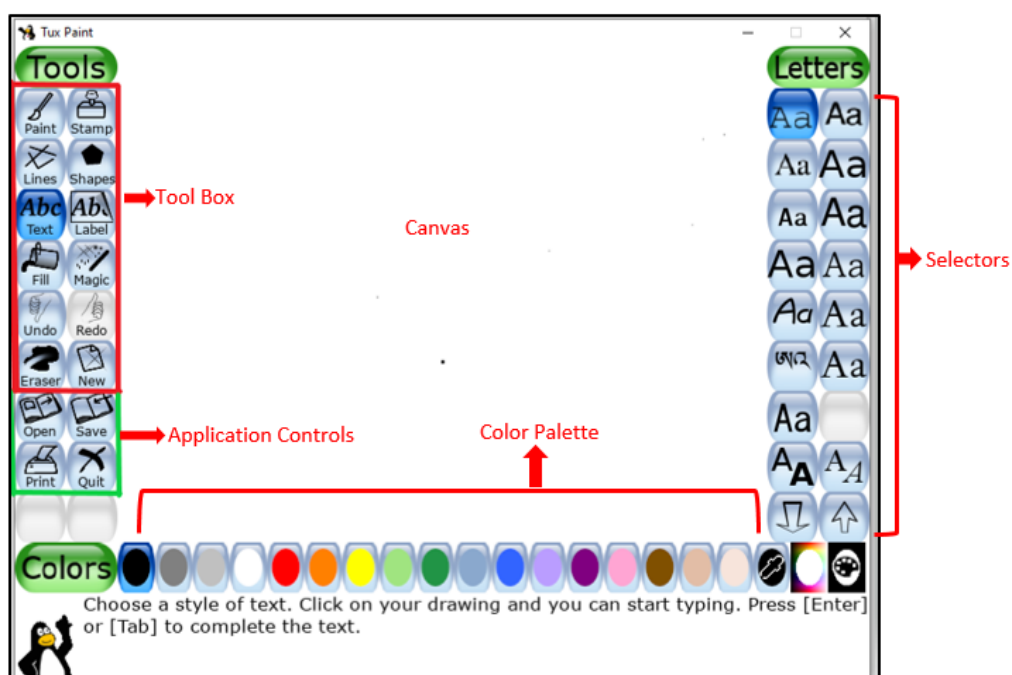
INTRODUCTION

Tux Paint is a free drawing program with a Graphics Editor. It has an easy interface, funny sound effects and drawing tools to make creative drawings. It has been developed by Bill Kendrick in the year 2002. You can download Tux Paint from the link:

<https://tuxpaint.org/download>

START WITH TUX PAINT

Click **Start** → **Tux Paint** → **Tux Paint (Windowed)**

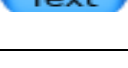
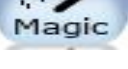


Components of Tux Paint

- ✓ **Tool Box:** Contains various basic tools. On the left side of the screen is the tool box, when any tool is clicked, on the right side of the screen all the options available under that tool are displayed.
Example- If you select Shapes tool then you can select different shapes from the right side to draw.
- ✓ **Application Controls:** Used for creating a new picture, save, undo and print them.
- ✓ **Canvas:** The area where images are drawn and edited.
- ✓ **Color Palettes:** You can select different colors to draw from here.
- ✓ **Selector:** Provides various selectable objects like brushes, fonts etc.

Drawing Tools of Tux Paint

Tux Paint has a large variety of Drawing Tools that help you to bring out your creativity.

TOOLS	NAME	DESCRIPTION
	Paint Brush	To create free hand drawing and readymade patterns.
	Stamp Tool	To add a readymade stamp in your drawing.
	Lines Tool	To draw straight lines and straight patterns.
	Fill Tool	Interactively fill an area with a solid color.
	Shape Tool	To draw polygonal shapes.
	Text Tool	To add text to the canvas.
	Magic Tool	To draw and give amazing effects to the picture.
	Eraser Tool	To erase the unwanted parts of the picture.

Drawing Pictures Using Paint Tool

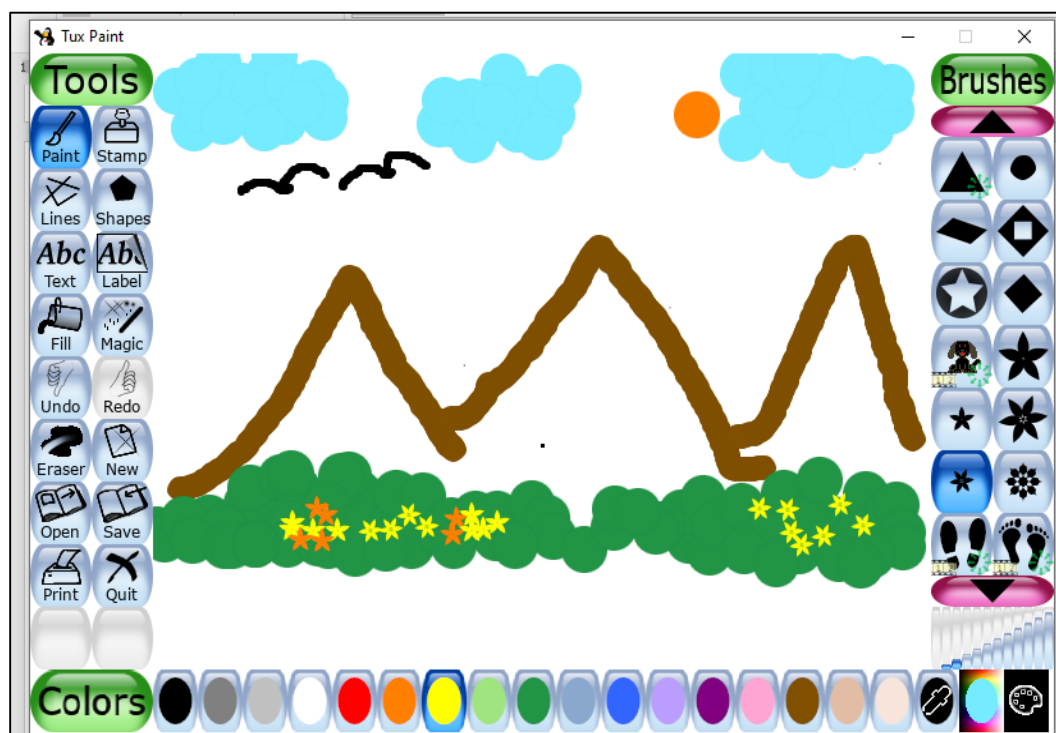
Step 1: Click on the **Paint Tool** in the Toolbar.

Step 2: Select any brush from the **Selector**.

Step 3: Choose a color from the **Color Palette**.

Step 4: Move the pointer to the Drawing Canvas. Left click at the position where you want to place the shape.

Repeat steps 1-4 with different brushes to draw a colorful picture as shown below.



Drawing Picture Using Lines and Eraser Tool

To draw pictures using Lines Tool

Step 1: Click on the **Lines Tool** from Toolbar.

Step 2: Click and choose a color from the **Colors Palette**.

Step 3: Click and select the size of the line from the **Selector**.

Step 4: Keeping the left mouse button pressed, move the mouse pointer on the Drawing Canvas. When you finish drawing, release the mouse.

To erase the picture using Eraser Tool

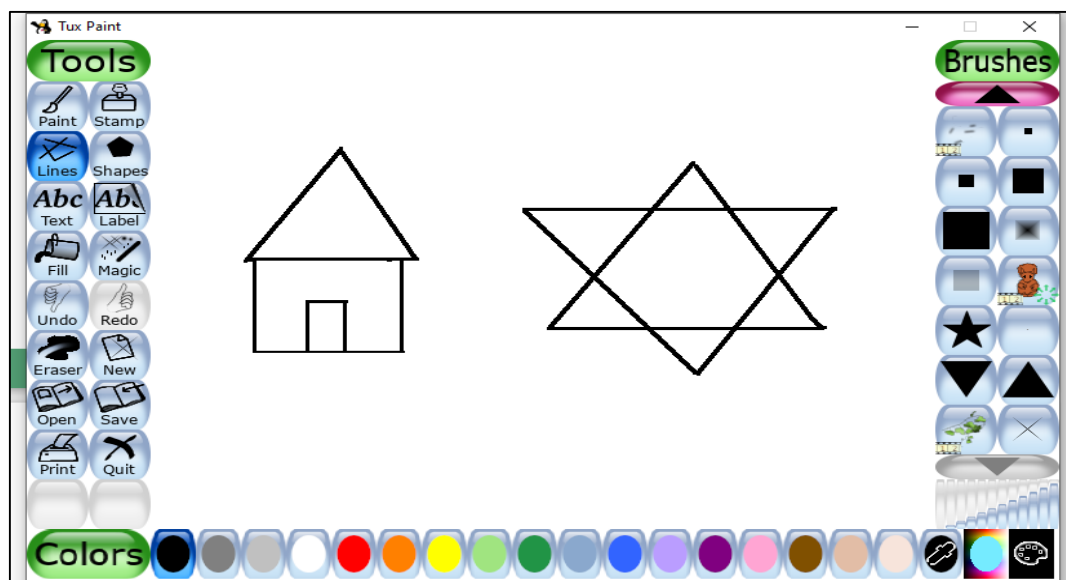
Step 1: Click on the **Eraser Tool** from Toolbar.

Step 2: Click and choose a size for the eraser from the **Selector**.

Step 3: Move the pointer to the Drawing Canvas. Keeping the left mouse button pressed, drag the mouse pointer over the part of the picture to be erased. When you finish erasing, release the mouse.

LAB ACTIVITY

- a) Draw the following pictures using the Lines tool.
- b) Erase the star from the canvas using Eraser Tool.



Drawing pictures using Shapes Tool

Step 1: Click on the **Shapes Tool** from Toolbar.

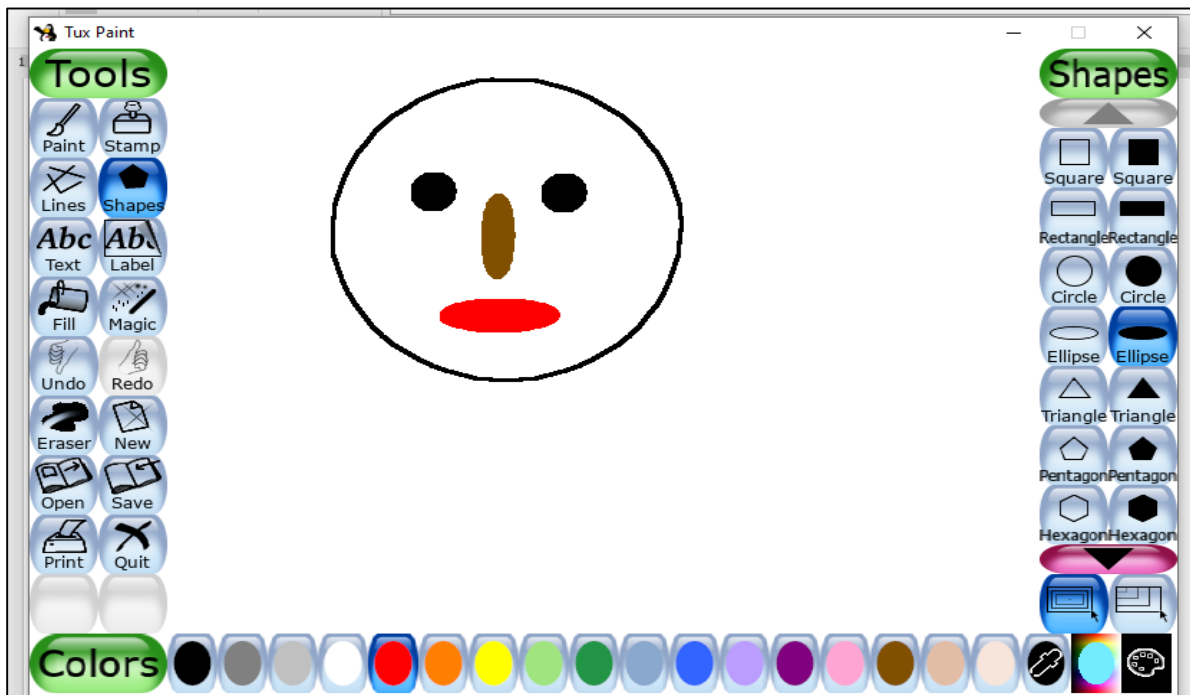
Step 2: Click and choose a color from the **Colors Palette**.

Step 3: Click and select a particular shape from the **Selector**.

Step 4: Keeping the left mouse button pressed, move the mouse pointer on the Drawing Canvas. When you finish drawing the shape, release the mouse.

LAB ACTIVITY

I. Draw the following picture using the Shapes Tool:



TEXT AND STAMP TOOL

Text Tool

The Text tool is used to type messages in the drawing canvas. Text entered using this tool can't be modified or moved later, since it becomes part of the drawing. However, because the text becomes part of the picture, it can be drawn over or modified using Magic tool effects (e.g., smudged, tinted, embossed, etc.)

To type text in your canvas:

Step 1: Click on the **Text Tool**.

Step 2: Click and choose a Letter Style from the **Selector**.

Step 3: Click and choose a color from the **Colors Palette**.

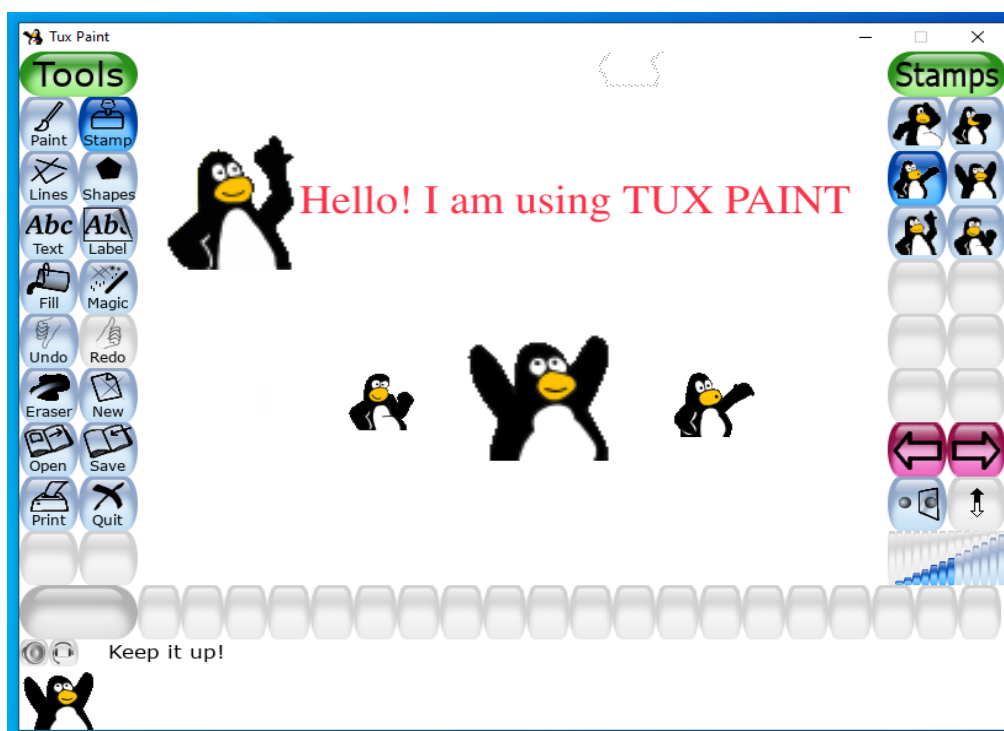
Step 4: Click on the Drawing Canvas. A blinking line called cursor appears. Type your message. Once you finish typing, click on any other tool in the Toolbar.

Stamp Tool

The Stamp tool works like a sticker or rubber stamp. It lets you place different pictures in the drawing. When this tool is selected, different stamps are displayed on the right side which can be used. Each stamp can be resized and flipped before being stamped. Resize button is found at the end of options and above the colors. This button can be used to increase or reduce the size of the stamps.

LAB ACTIVITY

- I. Draw the following using the Text and Stamp Tool:



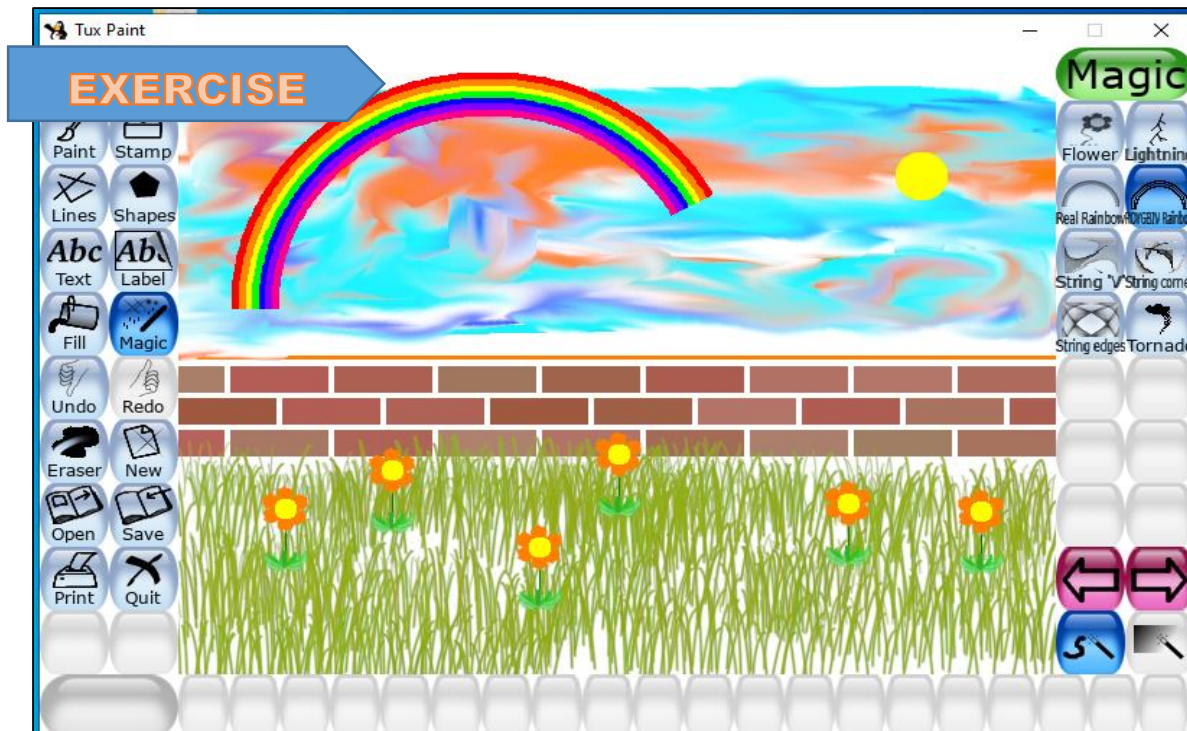
Magic Tool

The Magic tool is a set of many special tools. You can make different effects using this tool.

Step 1: Click on the **Magic Tool**.

Step 2: Click and choose a magic effect from the **Selector**.

Step 3: Move the pointer to the Drawing Canvas. Keep the left mouse button pressed, drag the mouse to draw on the Drawing Canvas. When you finish drawing release the mouse.



A. Choose the correct option.

- The developer of Tux Paint is _____ .
 a. Bill Gates ☐ b. Bill Kendrick ☐ c. Mark Zuckerberg ☐
- The _____ tool is used to enter text in your drawing .
 a. Text ☐ b. Line ☐ c. Paint ☐
- Which tool in Tux Paint is used to draw and give amazing effects to a picture?
 a. Paint ☐ b. Line ☐ c. Magic ☐
- Which among the following tools remove the unwanted part of drawing?
 a. Save ☐ b. Open ☐ c. Eraser ☐
- _____ tool is used to draw straight lines and straight patterns.
 a. Lines ☐ b. Text ☐ c. Paint ☐

B. Match the following:

- | | |
|--|---|
| 1.  | a) To add a readymade stamp in drawing. |
| 2.  | b) To draw polygon shapes. |
| 3.  | c) Interactively fill an area with a solid color. |
| 4.  | d) To type text to drawing. |
| 5.  | e) To draw straight lines and straight patterns. |

C. Rearrange the jumbled words:

- | | |
|------------------|-------|
| 1. NAVACS | _____ |
| 2. LROOC TEEATPL | _____ |
| 3. TORESLEC | _____ |
| 4. PAMST | _____ |
| 5. LOTO XOB | _____ |



CHAPTER

4

Presentation Tools

LEARNING OBJECTIVES:

At the end of this chapter, the students will be able to -

- Understand the concept of MS PowerPoint
- Create various slides for presentation
- Use Slide Layout in presentation
- Insert Text, Images and other objects in a slide
- Design Templates
- Save the Presentation
- Present the created slides using Slide Show

INTRODUCTION TO MS-POWERPOINT

PowerPoint is a presentation software program of the Microsoft Office package. PowerPoint uses a graphical approach to presentations in the form of slide shows that accompany the oral delivery of the topic. This program is widely used in business and classrooms and is an effective tool when used for training purposes.

A PowerPoint presentation is a collection of slides. A **Slide** is a frame or a page in a presentation that represents data. Presentations can contain text to display information or multimedia effects to make it impressive and interactive. Presentation files are saved with .pptx extension.

Advantages of using PowerPoint

- Presentation can be made to look very colorful and attractive.
- You can include text and picture movements on the screen.

DID YOU KNOW?

Approximately 35 Million PowerPoint Presentations are given each day by an estimated 500 million users worldwide.



GETTING STARTED WITH MS-POWERPOINT

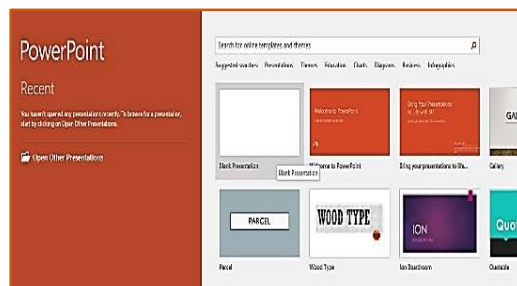
To start the MS-PowerPoint in your computer follow the steps below:

Step 1: Click on the Start button

Step 2: Type PowerPoint in the Search bar

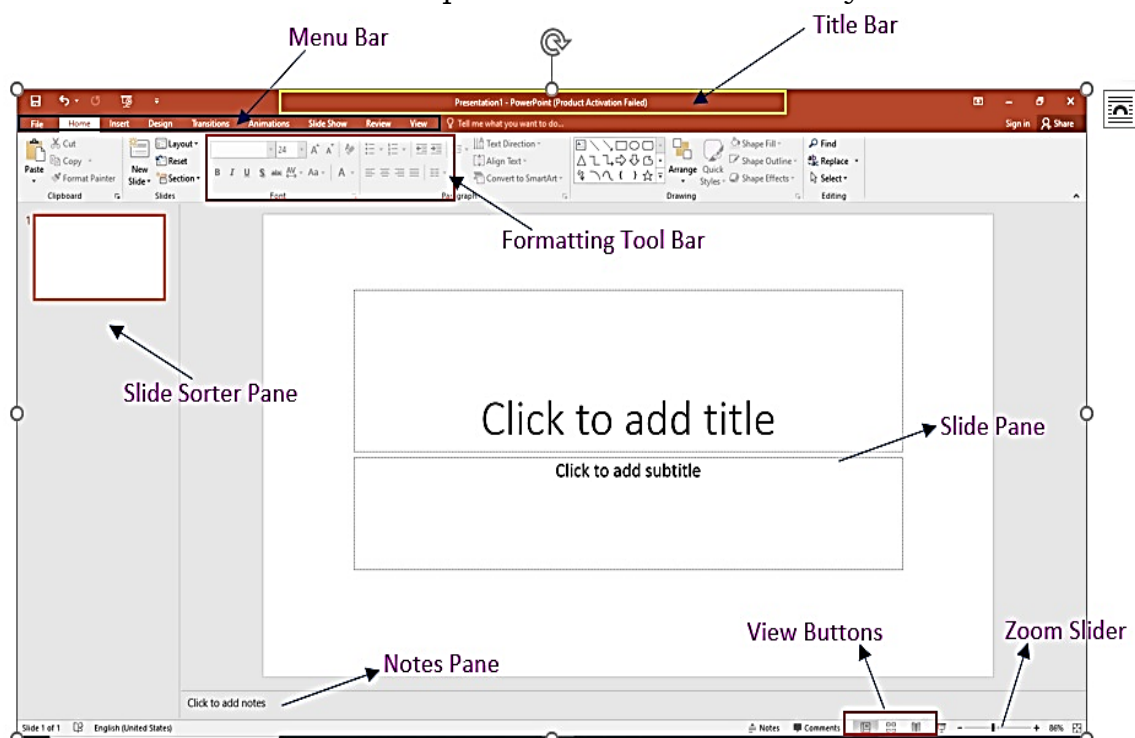
Step 3: Select the PowerPoint app to open the application

Step 4: Then click on Blank Presentation to create a new presentation or you can click on pre-installed templates and themes



Basic Elements of a PowerPoint Window/Slide

You can see various bars in the presentation window. They are:





- (i) **Title Bar:** This shows the name/title given by you to the current presentation. If user do not save the presentation by any name, default name given by MS PowerPoint appears in this bar.
- (ii) **Menu Bar:** Contains menu items like insert, views, design, animations, etc.
- (iii) **Formatting Toolbar:** Have tools like Bold, Italic, Underline, Font shape & size etc. to format your data.
- (iv) **Zoom Slider:** To zoom in or zoom out your presentation.
- (v) **Slide Sorter Pane:** This allows us to choose which slides will be shown in which sequence during the slide show.
- (vi) **Notes Pane:** This allows us to type notes that we may require later when preparing for the presentation, but they will not be displayed during the slide show.
- (vii) **View Buttons:** Provides different views of your presentation like: normal, slide show & slide sorter.
- (viii) **Slide Pane:** This is where we type, format, and otherwise design the slide.

Creating a New Slide

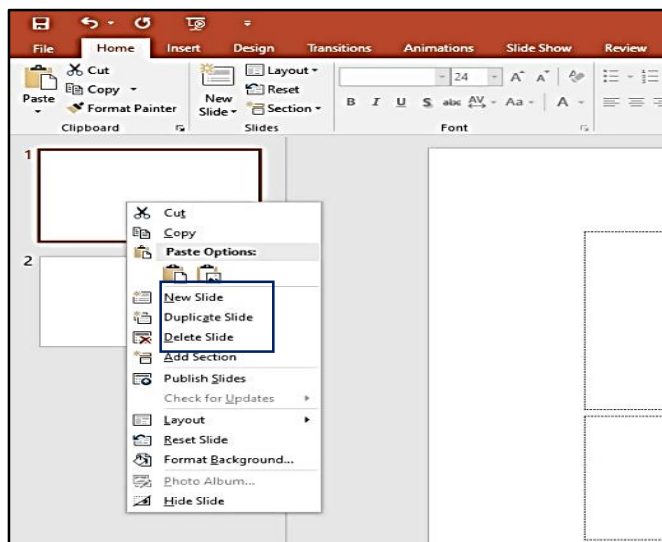
To create a new slide

1. Go to Home tab
2. Click on New Slide
3. Select the layout that you want for your new slide



Adding, Deleting, Duplicating & Rearranging the order of the Slides

1. To add slides, Right-click the slide in the thumbnail pane on the left, and select New Slide. (or press Ctrl + M)
2. To delete a slide, Right-click the slide in the thumbnail pane on the left, and select Delete Slide. (or click on the slide to be deleted and press delete button in the keyboard)



3. In the thumbnail pane on the left, right-click the slide thumbnail that you want to duplicate, and then click Duplicate Slide. The duplicate is inserted immediately after the original (or press Ctrl +D)
4. To Rearrange the order of slides, click the thumbnail of the slide that you want to move, and then drag it to the new location.

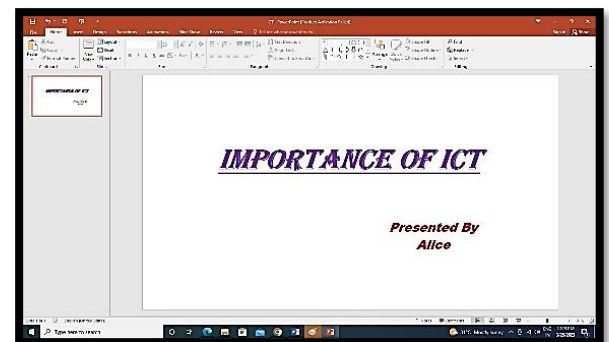
Creating a New Presentation

Step 1: Click in the title placeholder and type “Importance of ICT”.

Step 2: In the subtitle placeholder type “Presented by (your name).”

Step 3: Save the presentation with the name ICT.pptx (File-> Save).

Step 4: Close the presentation (File-> Close).

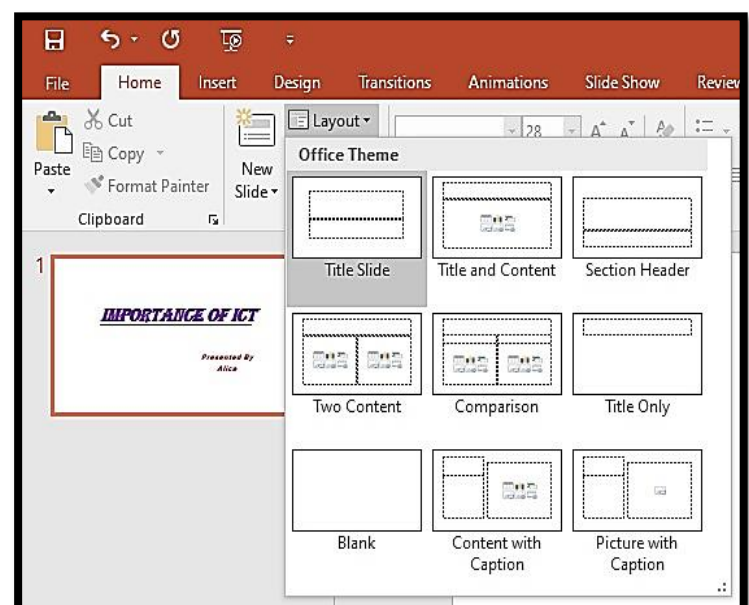


Slide layout

Layout is the method of placing things or objects on a slide in a specific order. A layout consists of placeholders or boxes with dotted borders which hold titles, body text, bulleted lists, etc.

Types of slide layout

- ❖ Title slide
- ❖ Section header
- ❖ Comparison
- ❖ Blank
- ❖ Title and content
- ❖ Two content
- ❖ Title only
- ❖ Content with caption
- ❖ Picture with caption



Inserting Text in a slide

Inserting text in a slide means adding text to a PowerPoint slide by typing inside the text box.

To insert a text in the Text Box

1. Go to Slide
2. Click on Text Box
3. Type the content

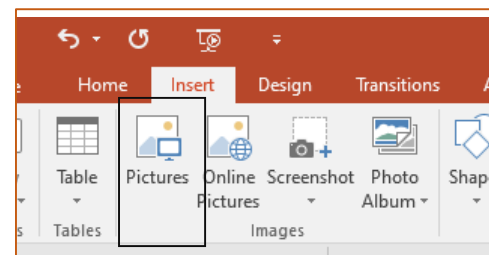


Inserting Images in a slide

Inserting image in a slide means adding picture to a PowerPoint slide.

To insert a image in a slide

1. Go to Insert
2. Click on Pictures
3. Select image from the location it is saved.

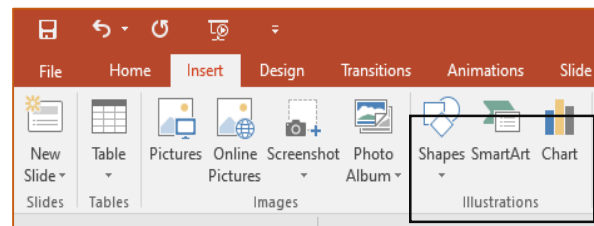


Inserting other objects like Shapes/SmartArt/Chart

Like images, other objects like shapes/smart art/chart can also be added to a PowerPoint slide.

To insert a other objects like Shapes/SmartArt/Chart

1. Go to Insert
2. Click on Shapes/SmartArt/Chart
3. Select the shape, smartart/chart you want in your presentation.



Design Templates

A PowerPoint template is a pattern or blueprint of a slide or group of slides. To add template:

1. Go to Design
2. Click to select the Template you want in your presentation



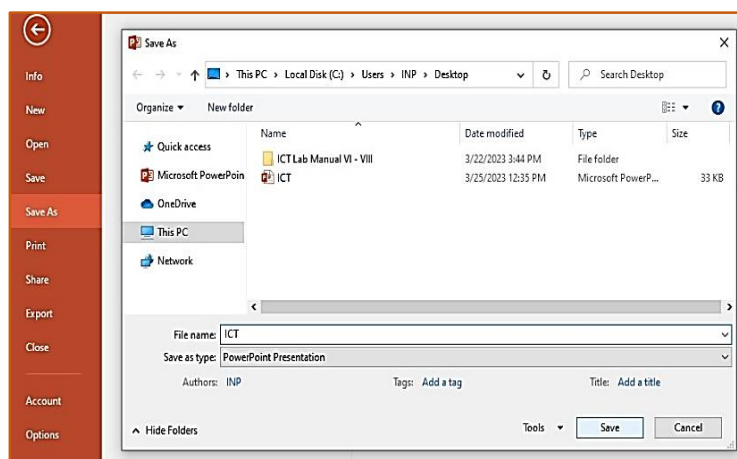
Saving a Presentation

You have created a slide now. It is advisable to save the presentation before adding more slides. The steps to save a presentation are:

1. Click on the File menu.
2. Select the Save option.

The Save As dialog box will appear

3. Type the name of the presentation "ICT" in the File Name box and choose location.
4. Click the Save button.



The presentation gets saved with an extension .pptx

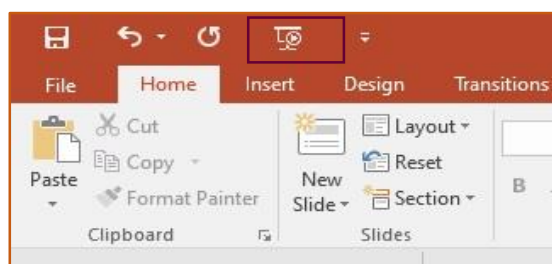
The Save As dialog box appears only the first time you save the file. When you choose the Save option again, PowerPoint will not ask you any questions and the latest changes will get saved. To save the file with different name or in other location, the Save As option is used under File Menu.

PRESENTING A SLIDE SHOW

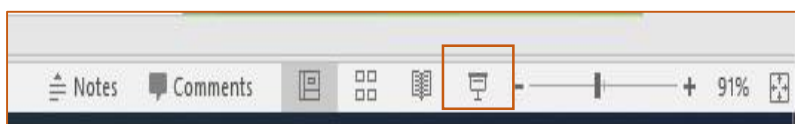
To start a slide show:

There are several ways you can begin your presentation:

- Click the **Start From Beginning** command on the Quick Access Toolbar, or press the **F5** key at the top of your keyboard. The presentation will appear in full-screen mode.



- Select the **Slide Show view** command at the bottom of the PowerPoint window to begin a presentation from the current slide.



- Go to the **Slide Show** tab on the Ribbon to access even more options. From here, you can start the presentation **from the current slide** and access **advanced presentation options**.



To advance and reverse slides:

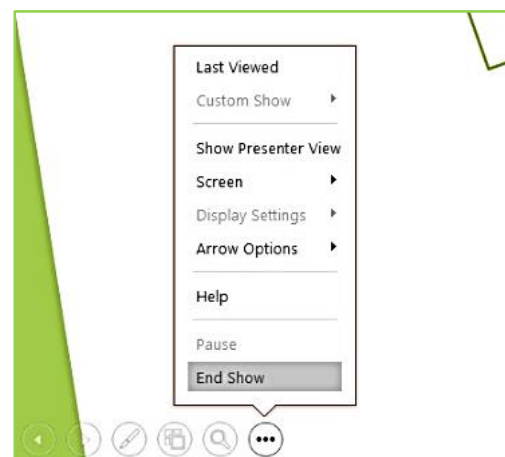
You can advance to the next slide by **clicking your mouse** or pressing the **spacebar** on your keyboard. Alternatively, you can use **arrow keys** on your keyboard to move forward or backward through the presentation.



You can also hover your mouse over the bottom-left and click the **arrows** to move forward or backward.

To stop a slide show:

You can exit presentation mode by pressing the **Esc** key on your keyboard. Alternatively, you can click the **Slide Show Options** button in the bottom-left and select **End Show**.



DO YOU KNOW?

PowerPoint was originally called "Presenter." It was created in 1987 by a company called Forethought Inc. for Macintosh. Later it was sold to Microsoft for \$14 Million.

ASSESSMENT

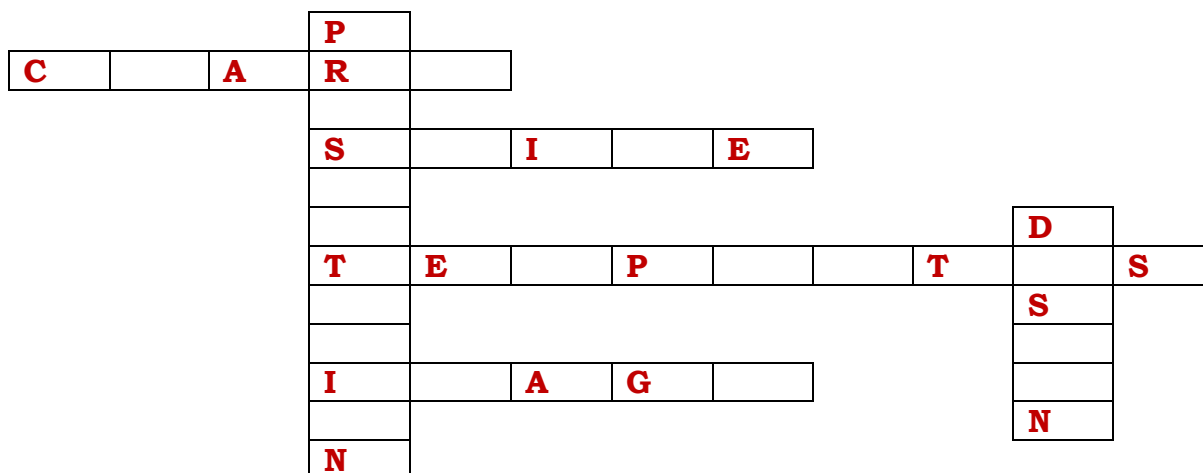
I. FILL IN THE BLANKS

- 1) Presentation files are saved with _____ extension.
- 2) _____ contains menu items like insert, views, design, animations, etc.
- 3) To insert a new slide, _____ command in the Slides group on the _____ tab should be clicked.
- 4) A PowerPoint _____ is a pattern or blueprint of a slide or group of slides.
- 5) _____ key from the keyboard is used to exit from the PowerPoint Presentation.

II. Match the following

Column A	Column B
Title Bar	To zoom in or zoom out your presentation
Formatting Toolbar	It shows the name/title given by you to the current presentation
Zoom Slider	It provides different views of your presentation
View Buttons	It has tools like Bold, Italic, Underline, Font shape & size etc.

III. Solve the crosswords using the words from the chapter.





CHAPTER

5

Communication & Network Terminologies

LEARNING OBJECTIVES:

At the end of this chapter, the students will be able to understand -

- **What is communication?**
- **Digital communication and its types-**
 - **Text & Images (SMS, MMS, Instant Messaging, Email), Phone calls, Video Conferencing**
- **Government platforms for ICT based Education**
- **Internet as Communication Tool**
- **Email & its features**
- **URLs**
- **Downloading content and image through Internet.**

WHAT IS COMMUNICATION?

Communication is a two-way process which involves transferring of information or messages from one person or group to another. This process goes on and includes a minimum of one sender and receiver to pass on the messages. These messages can either be any ideas, imagination, emotions, or thoughts.

Communication can be **Verbal** (written or oral communication with the help of languages) or **Non Verbal** (with the help of signs, symbols, colors, touches, body or facial features)

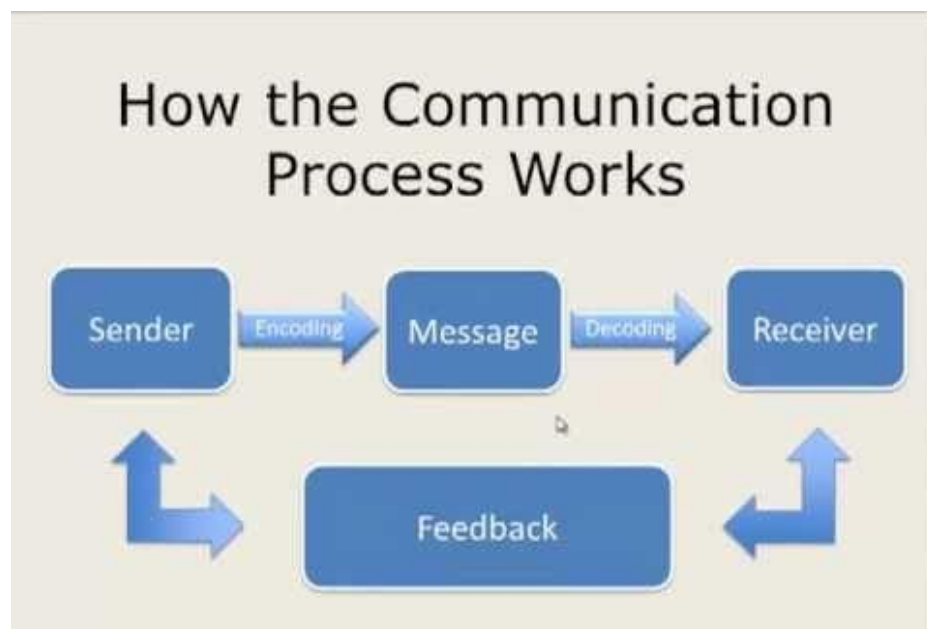
In the era of Information Technology, the means and modes of communications have also changed drastically. Letters are replaced by emails, mobile phones and other digital devices are taking place of landline phones etc.

WHAT IS DIGITAL COMMUNICATION?

Digital Communication refers to the process in which two or more digital devices are used to transfer data, instructions, and information. Some communications involve cables and wires; others are sent wirelessly through the air. Communications systems contain all types of computers and computing devices.

For successful communications, you need the following:

- A **sending device** that initiates an instruction to transmit data, instructions, or information.
- A communications device that connects the sending device to a communications channel. (e.g. modem)
- A **communications channel**, or transmission media on which the data, instructions, or information travel. (e.g. Cables)
- A communications device that connects the communications channel to a receiving device.
- A **receiving device** that accepts the transmission of data, instructions, or information.



Digital communication can be in the form of text, images (Non-Verbal Communication) or in the form of audio & video (Verbal Communication).



The following are some common forms of digital communications used in our day to day activities.

TYPES OF DIGITAL COMMUNICATION

SMS (Short Messaging Service)

SMS is a text messaging service component of most telephone, Internet and mobile device systems. The service allows users to send and receive messages of up to **160 characters** (including alpha-numeric and white space character).

MMS (Multimedia Messaging Service):

Multimedia Messaging Service (MMS) is a standard way to send messages that include multimedia content to and from a mobile phone over a cellular network. The MMS standard extends the core SMS (Short Message Service) capability, allowing the exchange of text messages greater than 160 characters in length. Unlike text-only SMS, MMS can deliver a variety of media, including up to forty seconds of video, one image, a slideshow of multiple images, or audio.

Instant Messaging(via the internet)

Unlike SMS, MMS, instant messaging uses the internet to send and receive messages. It is not limited by character count, but it does require both users to have same application. It also enables the users to send and receive text, images, audio, video, files and links. Some applications also support calls via internet E.g. Whatsapp Messenger.

E-mail:

Electronic mail (e-mail) is a computer-based application for the exchange of messages between users. A worldwide e-mail network allows people to exchange e-mail messages very quickly.

Phone Call: It is the verbal communication which is done by exchange of audio/voice over phone.



Video Conferencing:

Video conferencing is a visual communication session between two or more users regardless of their location, featuring audio and video content transmission in real time.

In broad terms, video conferencing software is a solution that allows two or more persons to talk and see each other remotely.

Common video conferencing Apps: Microsoft Teams, Google Meet, Zoom etc.



GOVERNMENT PLATFORMS FOR ICT BASED EDUCATION

The government of India is imagining and shaping E-learning with digital pedagogy as a long term strategy in the Education sector. It has contributed to the new age of learning which is not restricted to schools. Today the Ministry of Education is working upon the concept of learning by all, with all, and for all under the framework to adapt the enhanced learning experience for the students.

Prominent initiatives by Government for Digital Education:

1. DIKSHA (Digital Infrastructure for Knowledge Sharing)



DIKSHA (Digital Infrastructure for Knowledge Sharing) is a national platform for school education, an initiative of National Council for Educational Research and Training (NCERT). DIKSHA can be accessed by learners and teachers across the country and currently supports 36 Indian languages.



Some of its features are as follow:

- DIKSHA is the 'one nation; one digital platform' for school education in India.
- It was launched in 2017
- It is a national platform available for schools in all states
- DIKSHA is available for grades from 1 to 12.
- DIKSHA can be accessed through mobile application and web portal.

Web Portal : <https://diksha.gov.in>

or

You can download DIKSHA app for mobile phone via Google Play Store

2. Swayam Prabha – TV Channels

- This mode of education is for people who do not have access to formal education.
- High quality educational programmes are telecasted via television channels .
- There are a total of 32 channels to meet the requirements.
- Different channels are used for higher education and school education.
- The Department of School Education and Literacy also tied up with private DTH operators like Tata Sky & Airtel to air educational video content to enhance the reach of these channels.
- Number of TV channels for school education will increase from 5 to 12 to transform into 'one class, one channel', that is, one channel each for all grades from 1 to 12 channels.



3. e-Pathshala

- **e-Pathshala** mobile app (Android, iOS, Windows), and web portal can be used to access e-textbooks.
- It can be accessed by students, teachers and parents.
- 3,500 pieces of audio and video content of NCERT are available.





- It is available in different languages – English, Sanskrit, Urdu, Hindi.

4. Radio Broadcasting

1. The radio broadcasts focus on activity-based-learning.
2. For broadcasting content related to National Institute of Open Learning – NIOS (grades 9 to 12), 289 community radio stations have been used.
3. This mode of education is particularly useful for students who are living in remote areas, particularly for grades 1 to 5
4. Shiksha Vani is a Podcast of the Central Board for Secondary Education (CBSE)

INTERNET

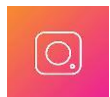
The Internet, sometimes called simply "the Net," is a worldwide system of computer networks - a network of networks in which users at any one computer can have access to information from other computers connected over the net without any geographical restrictions.

INTERNET AS A COMMUNICATION TOOL

Nowadays internet is being used as an effective means of communication both for personal and professional purposes like sending letters through email, posting on social media sites for wide publicity etc.

A list of various internet communication tools used in our day to day activities are as follow:

Instagram:



Instagram is a free photo and video sharing app available on iPhone and Android. People can upload photos or videos to our service and share them with their followers or with a select group of friends. They can also view, comment and like posts shared by their friends on Instagram.



Facebook

Facebook is a social networking site that makes it easy for an individual to connect and share with family and friends online. Facebook was created in 2004 by Mark Zuckerberg.



Twitter



Twitter is a free social networking site where users broadcast short posts known as tweets. These tweets can contain text, videos, photos or links. It is a microblogging service - a combination of blogging and instant messaging -- for registered users to post, share, like and reply to tweets with short messages. Non-registered users can only read tweets.

WhatsApp

WhatsApp is a free messaging app that lets users make video and voice calls, send text messages, share their status, and more with just a Wi-Fi connection.



Electronic mail (e-mail)



Electronic mail (e-mail) is a computer-based application for the exchange of messages between users. A worldwide e-mail network allows people to exchange e-mail messages very quickly. E-mail is the electronic equivalent of a letter, but with advantages in timeliness and flexibility.

YouTube



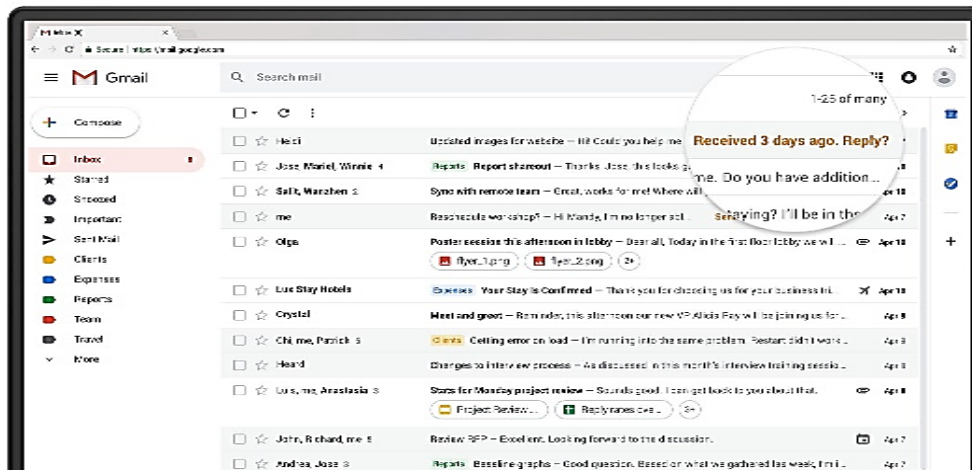
YouTube is a free video sharing website that makes it easy to watch online videos. We can even create and upload our own videos to share with others.

EMAIL & ITS FEATURES

Electronic mail, commonly shortened to “email,” is a communication method that uses electronic devices to deliver messages across computer networks.

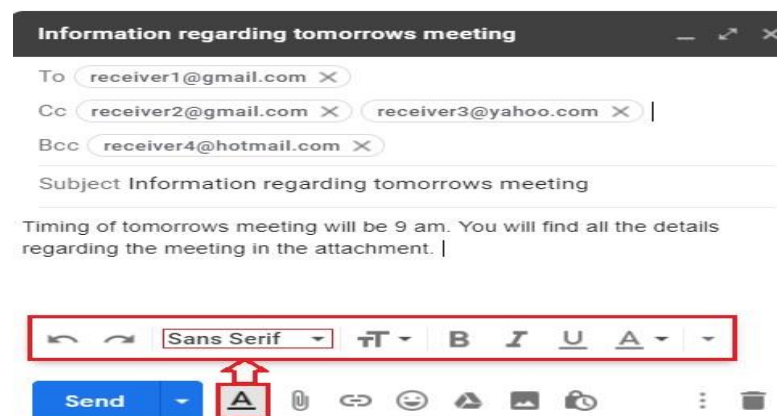


One of the main advantages of email is that you can quickly and easily send electronic files such as text documents, photos and data sheets to several contacts simultaneously. After you sign in, you will come to the home page where you can see different elements on the screen like Inbox, sent items etc.



Composing an Email

Once you create an email account, you can send emails to your friends. The process of sending an email is called Composing mail. When you click on compose button the following dialog box will be opened.



- **'To'** - an 'address bar' where you put the receiver's email address.
- **'Cc' (carbon copy)**- if you want to send the same message to other people and let every recipient to check who else has received the mail.
- **'BCc' (Blind Carbon Copy)** – No recipient in the To and Cc field can check who else has received the mail. Also, the recipient in the BCc will not be able to see the reply message else send privately.



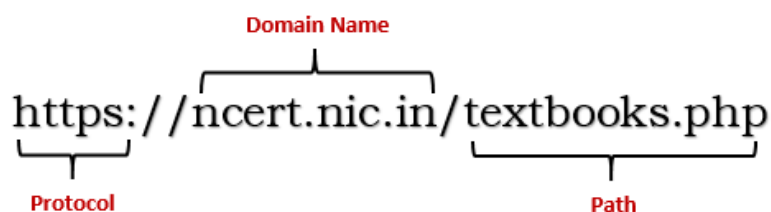
- **'Subject'** - a 'subject bar' is the title of your message, where you write what your email is about.
- The “Main Body” is divided into three sections.
 - Greetings/Salutations
 - A brief detail on the subject. It can be in the form of:
 - Single line
 - Minimum 3 para – introduction, body, conclusion (as required)
 - Email signature/sender’s details with contact information.
- **Send**- After typing your content, you can click on send button to

On the email page, you can see folders like these:

- **Inbox**: all the emails you have received.
- **Sent**: all the emails you have sent from this email address.
- **Draft**: email you have composed but not yet sent.
- **Trash**: emails which have been deleted before being deleted permanently after 30 days.
- **Spam**: unsolicited emails that are sent in bulk by companies or other groups.

URL (Uniform Resource Locator)

A URL (Uniform Resource Locator) is a unique identifier used to locate a resource on the Internet. It is also referred to as a web address. URLs consist of multiple parts - including a protocol and domain name (name of the website) that tell a web browser how and where to retrieve a specific page as file within the domain.





At first there were six main extensions:

- .in - India
- .com - Commercial use
- .net - Network / Internet service provider use
- .org - Organizational use
- .edu - Educational use (Schools and universities)
- .mil - Military use
- .gov - Governmental use

HOW TO DOWNLOAD A CONTENT AND IMAGE THROUGH INTERNET?

To download a file from remote server, perform the following steps:-

Step 1: Select the file/image/software you wish to download by clicking on the

 button or  (download) symbol or any download link.

Step 2: You might be prompted whether you wish to save or run the file from its present location. Select “Save” (if asked). [Some browser will choose “Save” automatically]

Step 3: A regular “Save As” dialog box appears, prompting to select the folder where to save the application or file. Select the path or save the file in Default path “C:\Downloads”.

Step 4: Click on save button. The file download will start.

Note: To download the image file we can also follow the below alternative steps:

1. Open the file to view it.
2. Right Click on the file.
3. Click on Download image or Save Image As option.
4. Select the path and change the default name.
5. Click on Save option.



I. Mark the following sentences as True or False.

1. Emails are slower than traditional mails. ☐
2. You don't need an email address to compose and send an email to someone. ☐
3. Trash is used for deleted emails. ☐
4. Draft is used for email which is not composed. ☐

II. Match the following:

A	B
i) Draft	Can be sent to more than one person.
ii) URL	Additional file in an email.
iii) Trash	Uniform Resource Locator.
iv) Email	Deleted Email.
v) Password	Email that is composed but not sent.
vi) Attachment	To protect and secure the account and data.

“Use the internet wisely for not just the petty enjoyment of your senses, but for the development of your mind as well.”

CHAPTER

6

Cyber Safety & Security – I

“While the internet is a fun place to meet people for gaming or chatting with friends, it’s also important to be cautious and to stay safe. This lesson will teach you some useful tips and tricks to help and keep yourselves safe.”

LEARNING OBJECTIVES:

At the end of this chapter, the students will be able to learn-

- ***What is Cyber?***
- ***Need of Cyber Security***
- ***Major Security problems***
- ***Virus & Worms***
- ***Hackers & Types of Hackers***
- ***Malware & Trojan Horses***
- ***Password cracking & Securing password***

CYBER

The word “Cyber” denotes a relationship with information Technology (IT) i.e. computers. (It can relate to all aspects of computing, including the storing data, protecting data, assessing data, processing data, transmitting data and linking data).



1. **Cyber-Crime** is wherein single actor or group of actors targets to harm individuals or systems for financial gains or to cause disruption
2. **Cyber-Attack** involves gathering information for political or economic means.
3. **Cyber-Terrorism** aims to undermine electronic systems with an intent to instil fear or to cause panic



Data is being misused by Cyber Criminals for various intentions like Geopolitical motivations, profit, steal, blackmail or extort, to disrupt communication, Ideological warfare, just for the sake of it, to be notorious, for personal gain, or to spread discontent

VIRUS AND WORMS

Virus

A computer virus is malicious code that replicates by copying itself to another program, computer boot sector or document and changes how a computer works.

A virus spreads between systems after some type of human intervention.



Some Sources of Computer Viruses

- Internet Viruses. An obvious way to get a computer infection is through the Internet such as malicious websites, malicious ads etc.
- Email - besides randomly downloading files on the Internet, email is the next most common source of threats such as scam links.
- Through Removable Storage devices.

- Through Bluetooth/wi-fi while sharing files from infected PC to another PC/phone.

Worms

A computer worm is a type of malware whose primary function is to self-replicate and infect other computers while remaining active on infected systems.

A computer worm duplicates itself to spread to uninfected computers.



Types of computer worms

- Email worms - Email worms work by creating and sending outbound messages to all the addresses in a user's contact list. ...
- File-sharing worms.
- Crypto-worms.
- Internet worms.
- Instant messaging worms.

HACKERS

A hacker is an individual who uses computer, networking or other skills to overcome a technical problem.

The term also may refer to anyone who uses their abilities to gain unauthorized access to systems or networks in order to commit crimes.



Types of Hackers

Hackers fall into three general categories.

1. Black Hat Hackers

- ❖ A black hat hacker is someone with objectives of studying and using cyber security techniques and tools for personal or private gain through malicious or threat activity.

2. White Hat Hackers

- ❖ A White Hat hacker tests systems and networks by trying to break into them.
- ❖ They are hackers, but their talents are being used to improve Cyber Security.

3. Gray Hat Hackers

- ❖ Gray hat hackers enact a blend of both black hat and white hat activities.
- ❖ Gray hat hackers often look for vulnerabilities in a system without the owner's permission or knowledge.
- ❖ If issues are found, they report them to the owner, sometimes requesting a small fee to fix the problem.



Hackers use malware to steal data directly from victims' computers using

- ❖ Spyware
- ❖ Remote administration malware
- ❖ Brute forced passwords
- ❖ Planting malware on victims' devices via infected websites
- ❖ Transmitting malicious attachments or links by email.

Although hackers are often associated with exploiting vulnerabilities to gain unauthorized access to computers, systems, or networks, not all hacking is malicious or illegal.

Malware

Malware is a catch-all term for any type of malicious software designed to harm or exploit any programmable device, service or network.

Cybercriminals typically use it to extract data that they can leverage over victims for financial gain.



Trojan Horses

A Trojan Horse (Trojan) is a type of malware that disguises itself as legitimate code or software. Once inside the network, attackers are able to carry out any action that a legitimate user could perform, such as exporting files, modifying data, deleting files or otherwise altering the contents of the device.

Password Cracking

Password cracking is the process of using an application program to identify an unknown or forgotten password to a computer or network resource.

It can also be used to help a threat actor obtain unauthorized access to resources.



Securing Password

Secure and strong passwords can help defend against cyberattacks and lower the risk of a security break.

- ❖ Password consist of a minimum of eight characters but preferably it must include 12 characters.
- ❖ It must include atleast one Uppercase letter alongwith lowercase letters and minimum one number and special character each.
- ❖ Strong passwords should not have any personal information.
- ❖ Security experts recommend using passphrases that combine several words and interchange numbers and symbols but are still easy to remember. Example : Passphrase – ‘Save Earth For Your Better Future’
Password – Se4ybf!!



Best ways to protect your password

- ❖ Never reveal your passwords to others.
- ❖ Use different passwords for different accounts.
- ❖ Use Multi-Factor Authentication (MFA).
- ❖ Length trumps complexity.
- ❖ Make passwords that are hard to guess but easy to remember.
- ❖ Complexity still counts.
- ❖ Use a password manager.

Think and answer!

- ✚ Cyber thieves might want to get access to your data or accounts, like your bank account. How would they do this?
- ✚ Cyber thieves might also want to steal your identity. Why would they do this?
- ✚ Cyber thieves might also want to meet you in person. Why would they do this?

Teacher's Note: Accept all related answers

ACTIVITY 1

Fill in the blanks

1. A strong password should contain at least one letter.
2. is the practice of sending an email which pretends to be from a company but is really trying to gather your personal?
3. A is a person who tries to steal your personal information online.
4. A fraudulent email is called
5. Your school computer system has to keep it secure.
6. A weak password is one that contains information.

ACTIVITY 2

True or False

1. Using a passphrase like a memorable song lyric or a string of random words is a good idea.
2. Using information that is easily identifiable to you is a bad idea when creating a password.
3. A password is strongest when you use only numbers.
4. Using symbols, such as '&' or '\$', can make your password even stronger.
5. Writing down your password in your school notebook is best because you will always have it handy.
6. Using different passwords for different apps and accounts will keep your data safer.



ACTIVITY 3

True or False

1. Any activity that seeks to compromise a digital device, computer system, or a network is defined as hacking.
2. Hacking is always illegal.
3. Malware and scams are examples of malicious cyber activity that can affect anyone.
4. All hackers are cyber criminals.
5. Hackers can be employed by companies and organisations to help make their data secure.

Solutions

Activity 1 (Fill in the blanks)

1. Capital
2. Phishing
3. Hacker
4. Spam
5. Antivirus
6. Personal

Activity 2 (True or False)

1. True
2. True
3. False
4. True
5. False
6. True

Activity 3 (True or False)

1. True
2. False
3. True
4. False
5. True



PROJECT

MS-Word, TUX Paint, Internet



Suggested activities:

1. Create a word document with at least 5 pages (teacher may give different topics to different students in group)
(Note: Students should use at least five of the following features in their document)
 - Use of Bullets and Numbering,
 - Insert Pictures,
 - Shapes and Symbols
 - Insert Page numbers, Headers, footers
 - Inserting Mathematical Equations and symbols in a document
 - Tables
2. Teacher may ask the students to create small images using various tools available in tux paint.
3. Teacher may ask students to download educational contents from the internet and save them in a folder with their name.
(Eg. Of Contents : NCERT textbooks from NCERT official Website , accessing video and audio content relevant to the school curriculum from DIKSHA etc.)

Note: 1. The teacher may assign the projects to students in groups of 3-4

2. The Project activities may be done in the ICT lab under the guidance of Teacher.



Akodara village of Gujarat situated in Sabarkantha district became the first digital village of India. ICICI Bank in 2015 under its Digital Village Project adopted Akodara village and made cashless by adopting digital technology.



STATE INSTITUTE OF EDUCATION

DEPARTMENT OF EDUCATION
ANDAMAN & NICOBAR ISLANDS